

Traceability Matrix

REMEMBER GORDON! – Rulebook $\rightleftharpoons$ Implementation Mapping

Generated from `docs/traceability.toml`

Overview

Implemented	Descriptive	Implicit	Out-of-scope
83	28	4	9

Total mappings: 124 · Total impl sites: 246

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0/3 implemented

§1 – Introduction

§1 – Introduction

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Introduction

§1.1 – General Comments

descriptive

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General Comments

“REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is a simulation of the final battle in Great Britain’s two-year campaign to reassert her presence in the Sudan (1896–1898). Fought September 2nd, 1898, Omdurman finally broke the back of the fanatical Dervish rebellion and gained Britain a million square miles of desolate territory and two million impoverished subjects. With two players, one assumes the role of Herbert Kitchener, Sirdar (CIC) of the Anglo-Egyptian army; the other player becomes the Khalifa, Abdullah the Taiasha, absolute ruler of the Dervishes. The game is also suited for multi-player participation, with each player assuming command of one or more Dervish tribes or Anglo-Egyptian brigades.

While “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is not, strictly speaking, a beginner’s game, the mechanics of play should be familiar to players of modest experience. It is suggested that the bonus game, FALL OF KHARTOUM, and the historical scenario be played first to familiarize players with the game system prior

(see manual for full text)

§1.2 – Game Scale

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Game Scale

Each hexagon of the mapsheet represents approximately 400–440 yards of real terrain and each day turn is the equivalent of two hours of real time. Each counter of infantry and cavalry represents between 400 and 700 men, and each of the gunboats present at the battle has its own counter. The upper echelon of command is represented by individual leader counters for the Anglo-Egyptian force; and leaders plus their retinues for the Dervish army.

 3/7 implemented

§2 – Game Components

§2 – Game Components

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Game Components

§2.1 – The Game Maps

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The Game Maps

The Omdurman battle map represents approximately 100 square miles of real territory and includes the area north of Omdurman in which the historical battle took place as well as the dominant terrain features that influenced the course of the battle. Note that the mapsheet also contains the Turn Record Track, Turn Sequence, and Terrain Effects Chart at the top; and the Combat Tables and Howitzer Fire Scattergram in the lower right corner. The large letters “A”, “D”, “Y”, etc. are set-up hexes for the historical scenario only (9.2) and should be ignored in the campaign game. Similarly, the hexsides of the Zariba exist only in the historical scenario and should be considered clear terrain in the campaign game. Note, however, that the Anglo-Egyptian player may “construct” the Zariba in the campaign game if desired (see 5.3). All full hexes of the Omdurman game map are playable, including the seven hexes of the Howitzer Fire Scattergram.

The mini-map for the bonus game, FALL OF KHARTOUM, represents that city as it appeared in 1885. The portion (see manual for full text)

See also: §5.3, §9.2

§2.2 – Play Aids

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Play Aids

Certain charts and tables are needed to play the game. The Terrain Effects Chart lists all terrain found on the mapsheet and the effect of each type on movement and combat. The Combat Tables describe the range effects on various weapon types and includes the Combat Results Table. Also note the Line of Sight Table on the back of this rulebook. It tells players when certain terrain types block line of sight, thus preventing direct fire attacks on enemy units. Players should become familiar with these various charts and tables prior to the beginning of play.

(see manual for full text)

§2.3 – The Units

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The Units

File	Symbol	Code Snippet
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omdurman-types/src/lib.rs:842 GH:omdurman-types/src/lib.rs:842	UnitKind	<pre> 840 /// `Some(UnitKind::Marker)` or `None`. 841 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 842 pub enum UnitKind { 843 /// Foot infantry (\$2.3): fire / melee / movement. 844 Infantry { </pre>
omdurman-rules/src/lib.rs:728 GH:omdurman-rules/src/lib.rs:728	UnitProfile	<pre> 726 /// print no melee value). 727 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 728 pub struct UnitProfile { 729 pub kind: UnitKind, 730 pub identity: UnitIdentity, </pre>
omdurman-rules/src/lib.rs:16 GH:omdurman-rules/src/lib.rs:16	BrigadeId	<pre> 14 15 use omdurman_types::{ 16 BrigadeId, BrigadeNationality, DayNight, DervishTribe, Faction, HexCoord, HexsideRef, Player, 17 SetupLetter, UnitKind, 18 }; </pre>
omdurman-types/src/lib.rs:657 GH:omdurman-types/src/lib.rs:657	SpriteAnnotation	<pre> 655 /// as an optional overlay over the compiled `sprite_data` fallback. 656 #[derive(Serialize, Deserialize, Clone, Debug, PartialEq)] 657 pub struct SpriteAnnotation { 658 pub color: SpriteColor, 659 pub faction: Option<Faction>, </pre>

Covered by tests: [omdurman-rules::src::unit_profiles::british_army_row_zero_specials_classify_by_counter](#)
[omdurman-rules::src::unit_profiles::egyptian_army_row_zero_specials_classify_by_counter](#)

§2.4 – Game Parts Inventory

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Game Parts Inventory

Your complete copy of “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN includes:

- One 22 × 28 Battle of Omdurman mapsheet
- One 8½ × 11 bonus game: FALL OF KHARTOUM mapsheet
- One Rules Booklet
- One die-cut Unit Counter Sheet
- One Campaign Game Order of Appearance Card
- One ten-sided die
- One game box

§2.31 – Dervish weapon types

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Dervish artillery, gunboats, and forts fire on the “artillery” line of the Dervish Range Effects Table; Jehadia and Danagla units fire on the “rifles” line as does the Isa Zachneih unit. All other Dervish units (including leaders) are armed with spears and swords.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:460	WeaponClass	<pre> 458 /// enum so a "spear" unit cannot accidentally fire on the "Howitzer" line. 459 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] </pre>

GH:omdurman-rules/src/lib.rs:460		<pre> 460 pub enum WeaponClass { 461 /// Dervish spears and swords -- no ranged fire at all. 462 Melee, </pre>
omdurman-rules/src/unit_profiles.rs:510 GH:omdurman-rules/src/unit_profiles.rs:510	dervish_tribe	<pre> 508 /// Resolve a Dervish tribal foot counter (§2.31): Jehadia, Danagla and 509 /// Isa Zachneih fire on the rifles line; every other tribe is spear-armed. 510 pub fn dervish_tribe(tribe: DervishTribe) -> Option<Classification> { 511 // §2.31: "Jehadia and Danagla units fire on the 'rifles' line as does the 512 // Isa Zachneih unit. All other Dervish units (including leaders) are armed </pre>
omdurman-rules/src/unit_profiles.rs:292 GH:omdurman-rules/src/unit_profiles.rs:292	khalifa_abdullah	<pre> 290 /// battle (§9.322). All three are interchangeable, so they share the 291 /// `DervishArtillery` identity. 292 pub fn khalifa_abdullah(col: u32, row: u32) -> Option<Classification> { 293 let artillery = { 294 Some(Classification { </pre>

Covered by tests: [omdurman-rules::src::unit_profiles::dervish_weapon_class_follows_the_rifles_line](#)

§2.32 – Anglo-Egyptian weapon types

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All Anglo-Egyptian units (except gunboats, Maxims, artillery, and leaders) are armed with rifles. Maxims fire on the “Maxims” line of the Anglo-Egyptian Range Effects Table, and artillery and old gunboats fire on the “Artillery” line. New type (named) gunboats may fire on the “Howitzer”, “Artillery”, and “Maxims” lines of the Range Effects Table. (See 6.52 for the fire capabilities of the “Friendlies”).

****Sample Dervish Units**** (printed on counters): combat unit (Combat / Melee / Movement values, plus Tribe identifier); Leader (e.g. OSMAN DIGNA); Camel unit (e.g. Danagla, 4-6-12); Fort.

****Sample Anglo-Egyptian Units**** (printed on counters): Cavalry (e.g. 21 Lancers); Artillery (e.g. 32 Battery); Old Gunboat (e.g. LORD KITCHENER, 0-0-15); New Gunboat — named (e.g. Sultan, with artillery and howitzer factor, plus movement downstream / movement upstream values); Maxim Guns (fire twice per turn); Infantry (Fire Combat Factor / Melee / Movement, plus Battalion ID and Brigade ID — e.g. “2B” = 2nd British Brigade, “3E” = 3rd Egyptian Brigade).

(see manual for full text)

See also: §6.52

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:414 GH:omdurman-rules/src/lib.rs:414	GunboatId	<pre> 412 /// fire; "old" gunboats do not (rulebook §2.32). 413 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 414 pub enum GunboatId { 415 /// One of the five new-type named gunboats with howitzer capability. 416 Named(NamedGunboat), </pre>
omdurman-rules/src/lib.rs:433 GH:omdurman-rules/src/lib.rs:433	NamedGunboat	<pre> 431 /// The five named gunboats with howitzer capability (rulebook §6.64, §2.32). 432 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 433 pub enum NamedGunboat { 434 Sultan, 435 Melik, </pre>
omdurman-rules/src/lib.rs:446	OldGunboat	<pre> 444 /// in the Maxim Second Fire and Howitzer subphase (§6.42). </pre>

GH:omdurman-rules/src/lib.rs:446		445 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 446 pub enum OldGunboat { 447 LordKitchener, 448 Tamai,
omdurman-rules/src/lib.rs:446 GH:omdurman-rules/src/lib.rs:446	GunboatId::Old	444 /// in the Maxim Second Fire and Howitzer subphase (\$6.42). 445 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 446 pub enum OldGunboat { 447 LordKitchener, 448 Tamai,
omdurman-rules/src/lib.rs:420 GH:omdurman-rules/src/lib.rs:420	GunboatId::Dervish-Gunboat	418 Old(OldGunboat), 419 /// A Dervish gunboat (\$9.111, \$10.14). 420 DervishGunboat(u8), 421 } 422

Covered by tests: [omdurman-rules::src::effects::old_gunboat_lacks_howitzer](#)
[omdurman-rules::src::effects::old_gunboat_rejected_from_howitzer_subphase](#)

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| §3 – Getting Started

§3 – Getting Started

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Getting Started

Spread out the mapsheet on a table. It should lie flat if you backfold it against the scored lines. The Dervish player should sit next to the west edge of the map and the Anglo-Egyptian player opposite him on the east edge. Read through the rules once, looking over the various charts as they are referred to in the various sections. Next, select a scenario and punch out only those unit counters needed to play. Later on, the rest of the unit counters should be punched out, sorted and stored by unit type.

(see manual for full text)

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§4 – Turn Sequence

§4 – Turn Sequence

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Turn Sequence

“REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is played in “Game Turns”, each of which has two “Player Turns”. The player moving first will vary according to the scenario being played. In the Campaign Game, for example, the Anglo-Egyptian player moves first.

A) Anglo-Egyptian Player Turn:

1. Anglo-Egyptian Movement Phase
2. Fire Combat Phase a. Dervish Defensive Fire b. Anglo-Egyptian Offensive Fire
 1. Direct Fire Subphase
 2. Maxim Second Fire and Howitzer Fire Subphase

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:238 GH:omdurman-rules/src/lib.rs:238	GameTurnIndex	<pre>236 /// One-based Game Turn index (1, 2, ... up to the 237 scenario length) (rulebook §4). 237 #[derive(Serialize, Deserialize, Clone, Copy, 238 PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 238 pub struct GameTurnIndex(u8); 239 240 impl GameTurnIndex {</pre>
omdurman-rules/src/lib.rs:260 GH:omdurman-rules/src/lib.rs:260	Phase	<pre>258 /// etc. 259 #[derive(Serialize, Deserialize, Clone, Copy, 260 PartialEq, Eq, Debug, Default)] 260 pub enum Phase { 261 /// Pre-game deployment (§9.2/§9.3/§10): fixed 262 units are placed, each side 262 /// deploys its order of battle within its legal 262 zone, and river</pre>
omdurman-rules/src/effects.rs:823 GH:omdurman-rules/src/effects.rs:823	GameState	<pre>821 /// All mutable state of a game in progress (rulebook 822 §4). 822 #[derive(Serialize, Deserialize, Clone, Debug)] 823 pub struct GameState { 824 pub scenario: Scenario, 825 pub current_turn: GameTurnIndex,</pre>
omdurman-rules/src/effects.rs:959 GH:omdurman-rules/src/effects.rs:959	GameState::new	<pre>957 impl GameState { 958 /// Create a fresh game state for a given scenario 959 (rulebook §4). 959 pub fn new(scenario: Scenario) -> Self { 960 let first = scenario_turn(scenario, 961 GameTurnIndex::new(1)); 961 // First player to *move* per scenario: 961 Campaign -- Anglo-Egyptian moves</pre>
omdurman-rules/src/effects.rs:50 GH:omdurman-rules/src/effects.rs:50	AdvancePhase	<pre>48 /// At end-of-turn, disrupted units recover and 49 per-turn tracking is 49 /// cleared. 50 AdvancePhase, 51 52 // -- Movement -----</pre>
omdurman-rules/src/effects.rs:2745 GH:omdurman-	advance_phase	<pre>2743 2744 /// Advance the game state to the next phase (rulebook 2745 §4).</pre>

rules/src/ effects.rs:2745		<pre> 2745 pub fn advance_phase(state: &mut GameState) -> Result<(), RuleError> { 2746 let old_phase = state.phase; 2747 debug!(old_phase = ?old_phase, active_player = ? state.active_player, "advance_phase"); </pre>
omdurman- rules/src/ effects.rs:2839 GH:omdurman- rules/src/ effects.rs:2839	end_player_turn	<pre> 2837 2838 /// End the current player's turn: recover disrupted units, switch active player, advance turn index (rulebook §4). 2839 pub fn end_player_turn(state: &mut GameState) -> Result<(), RuleError> { 2840 debug!(2841 old_player = ?state.active_player, </pre>
omdurman- rules/src/ lib.rs:57 GH:omdurman- rules/src/lib.rs:57	GameTurnIn- dex::value	<pre> 55 56 impl \$name { 57 pub fn value(self) -> u16 { 58 match self { 59 \$(Self::\$variant => \$value,)+ </pre>
omdurman- rules/src/ effects.rs:951 GH:omdurman- rules/src/ effects.rs:951	PendingMelee	<pre> 949 /// resolution after the reaction window is deterministic and host-ordered (rulebook §7.5). 950 #[derive(Serialize, Deserialize, Clone, Debug)] 951 pub struct PendingMelee { 952 pub attack: MeleeAttack, 953 pub attacker_roll: DieRoll, </pre>

Covered by tests:

- omdurman-rules::src::effects::both_ready_auto_advances_out_of_setup
- omdurman-rules::src::effects::fire_combat_wrong_phase_rejected
- omdurman-rules::src::effects::new_game_starts_in_setup
- omdurman-rules::src::turn_track::scenario_turn_dispatches_correctly
- omdurman-rules::src::effects::turn_advances_through_phases
- omdurman-rules::src::unit_profiles::game_turn_marker_cell_returns_none

18/23 implemented

§5 – Movement Phase

§5 – Movement Phase (general)

descriptive

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Movement Phase

Covered by tests: `omdurman-rules::src::effects::disrupted_unit_cannot_fire`

§5.1 – General Rules

descriptive

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General Rules

§5.2 – Movement Restrictions

descriptive

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Movement Restrictions

§5.3 – Constructing the Zariba

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Constructing the Zariba

The Zariba trench and thorn hedge hexsides are built and in place in the historical scenario only. These hexsides are considered clear terrain in the campaign game. The Anglo-Egyptian player may, however, find it useful to construct this defensive position during the campaign game. The Zariba hexsides may only be built in their position as displayed on the mapsheet. Construction procedure is as follows: any Anglo-Egyptian infantry unit that begins and ends the Anglo-Egyptian player turn adjacent to (and on the Nile side of) Zariba hexsides has constructed all Zariba hexsides to which he is adjacent. The constructing unit may neither fire offensively nor melee attack during the turn of construction. Use a blank counter to denote units constructing Zariba hexsides. See 9.23 for defensive benefits and movement restrictions of Zariba hexsides.

(see manual for full text)

See also: §9.23

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:762 GH:omdurman-rules/src/lib.rs:762	constructing_zariba	<pre>760 /// Set while the unit is building Zariba hexsides -- neither offensive 761 /// fire nor melee allowed that turn (§5.3). 762 pub constructing_zariba: bool, 763 /// Set when the Royal Engineers are committed to a demolition this turn 764 /// (§6.53) -- neither offensive fire nor melee allowed that turn.</pre>
omdurman-rules/src/effects.rs:202	ConstructZariba	<pre>200 201 /// Begin constructing a Zariba hexside (rulebook §5.3).</pre>

GH:omdurman-rules/src/effects.rs:202		202 ConstructZariba { 203 unit_ids: Vec<UnitId>, 204 hexside: HexsideRef,
omdurman-rules/src/effects.rs:4571 GH:omdurman-rules/src/effects.rs:4571	apply_construct_zariba	4569 4570 /// Mark a set of units as constructing a Zariba hexside (rulebook §5.3). 4571 pub fn apply_construct_zariba(4572 state: &mut GameState, 4573 unit_ids: &[UnitId],

Covered by tests: [omdurman-rules::src::effects::construct_zariba_marks_builders_and_records_hexside](#)

§5.4 – Zones of Control

[descriptive](#)

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Zones of Control

§5.5 – Stacking

[descriptive](#)

manual page unknown

Stacking

§5.11 – Movement allowances printed on units

[implemented](#)

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The movement allowances of the various unit types are printed directly on the units (see 2.3). A unit may move up to this printed movement allowance, paying varying costs for different terrain types (see the Terrain Effects Chart).

See also: §2.3

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:125 GH:omdurman-rules/src/lib.rs:125	MovementAllowance	123 // is a named variant. 124 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 125 pub enum MovementAllowance { 126 // Immobile (forts, wrecked gunboats). 127 Immobile = 0,
omdurman-rules/src/lib.rs:739 GH:omdurman-rules/src/lib.rs:739	UnitMovement	737 // Movement allowance -- uniform for land units, split for gunboats (rulebook §5.11, §5.24, §5.25). 738 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 739 pub enum UnitMovement { 740 Land(MovementAllowance), 741 Gunboat(GunboatMovement),
omdurman-types/src/lib.rs:280 GH:omdurman-types/src/lib.rs:280	HexDirection	278 // (`+q`, `+q+r`, `+r`, `-q`, `-q-r`, `-r` for pointy-top hexes) (rulebook §5.11, §5.24). 279 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, Default)] 280 pub enum HexDirection { 281 #[default] 282 East = 0,
omdurman-rules/src/	MovementPoints	165 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug, Default,

lib.rs:167 GH:omdurman-rules/src/lib.rs:167		166 <code>)]</code> 167 <code>pub struct MovementPoints(i16);</code> 168 169 <code>impl MovementPoints {</code>
omdurman-rules/src/terrain_chart.rs:21 GH:omdurman-rules/src/terrain_chart.rs:21	terrain_effects_chart	19 <code>///</code> 20 <code>/// Source: printed Terrain Effects Chart on the mapsheet.</code> 21 <code>pub fn terrain_effects_chart(terrain: Terrain) -></code> TerrainEntry { 22 <code> match terrain {</code> 23 <code> Terrain::Clear { .. } => TerrainEntry {</code>
omdurman-rules/src/terrain_chart.rs:77 GH:omdurman-rules/src/terrain_chart.rs:77	movement_cost	75 <code>/// Convenience: get the movement cost for a terrain type (rulebook §5.11, Terrain Effects Chart).</code> 76 <code>/// Returns `None` for impassable terrain (Nile).</code> 77 <code>pub fn movement_cost(terrain: Terrain) -></code> Option<MovementAllowance> { 78 <code> terrain_effects_chart(terrain).movement_cost</code> 79 <code>}</code>
omdurman-rules/src/terrain_chart.rs:85 GH:omdurman-rules/src/terrain_chart.rs:85	movement_cost_with_road	83 <code>/// underlying terrain; without a road it's the terrain's own cost. The road is</code> 84 <code>/// a movement overlay only -- combat/LOS still use the underlying terrain.</code> 85 <code>pub fn movement_cost_with_road(terrain: Terrain, road:</code> bool) -> Option<MovementAllowance> { 86 <code> if road {</code> 87 <code> Some(MovementAllowance::One)</code>
omdurman-types/src/lib.rs:476 GH:omdurman-types/src/lib.rs:476	Terrain::has_road	474 475 <code>/// Whether this hex has any road touching it.</code> 476 <code>pub fn has_road(self) -> bool {</code> 477 <code> !matches!(self.road(), Road::None)</code> 478 <code>}</code>
omdurman-types/src/lib.rs:413 GH:omdurman-types/src/lib.rs:413	Terrain::passable_by_land	411 412 <code>/// Whether this terrain may be entered by land units (rulebook §5.11).</code> 413 <code>pub fn passable_by_land(self) -> bool {</code> 414 <code> !self.is_nile()</code> 415 <code>}</code>
omdurman-types/src/lib.rs:481 GH:omdurman-types/src/lib.rs:481	Terrain::is_crossroad	479 480 <code>/// Whether roads converge at this hex's centre.</code> 481 <code>pub fn is_crossroad(self) -> bool {</code> 482 <code> matches!(self.road(), Road::Crossroad)</code> 483 <code>}</code>
omdurman-hexmap/src/map.rs:17 GH:omdurman-hexmap/src/map.rs:17	GameMap::roads	15 <code>pub hexes: HashMap<HexCoord, HexData>,
16 <code>pub hexsides: HashMap<HexsideRef, HexsideKind>,
17 <code>pub roads: HashSet<HexsideRef>,
18 <code>pub excluded: HashSet<HexCoord>,
19 <code>pub overlay: OverlayParams,</code></code></code></code></code>

Covered by tests: `omdurman-rules::src::terrain_chart::clear_terrain_no_bonus` `omdurman-rules::src::terrain_chart::nile_is_impassable`
`omdurman-rules::src::terrain_chart::rough_movement_and_defense`
`omdurman-rules::src::terrain_chart::swamp_movement_and_defense`
`omdurman-rules::src::terrain_chart::hilltop_movement_and_defense`
`omdurman-rules::src::terrain_chart::huts_movement_and_defense`
`omdurman-rules::src::terrain_chart::movement_cost_convenience_matches_chart`
`omdurman-rules::src::terrain_chart::movement_cost_with_road_always_one`
`omdurman-rules::src::effects::land_unit_may_not_enter_nile`
`omdurman-rules::src::terrain_chart::movement_cost_without_road_matches_terrain`
`omdurman-rules::src::effects::movement_cost_for_uses_terrain` `omdurman-rules::src::effects::movement_cost_for_road_costs_one`
`omdurman-rules::src::terrain_chart::road_gives_crossroad` `omdurman-rules::src::terrain_chart::terrain_movement_costs_in_bounds`
`omdurman-rules::src::terrain_chart::terrain_chart_road_always_costs_one`

§5.12 – Move up to allowance, hex by hex (cumulative MP cap)

implemented

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A player may move as many or as few of his units as desired during each movement phase, limited only by the units' movement allowance, the terrain costs paid in moving from hex to hex, and enemy zones of control (see 5.4).

See also: §5.4

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1483 GH:omdurman-rules/src/effects.rs:1483	can_move_unit	<pre>1481 /// the same `RuleError` the `MoveUnit` effect 1482 would on rejection. Lets the 1483 /// UI gate input without mutating or duplicating 1484 the rules. 1485 pub fn can_move_unit(&self, unit_id: UnitId, cost: MovementPoints) -> Result<(), RuleError> { 1486 self.can_move_unit_to(unit_id, None, cost) 1487 }</pre>
omdurman-rules/src/effects.rs:2174 GH:omdurman-rules/src/effects.rs:2174	mp_spent	<pre>2172 /// Movement points `unit_id` has already spent 2173 this turn (§5.11/§5.12). 2174 pub fn mp_spent(&self, unit_id: UnitId) -> i16 { 2175 self.mp_spent_this_turn.get(&unit_id).copied().unwrap_or(0) 2176 }</pre>

Covered by tests: [omdurman-rules::src::effects::cumulative_move_cost_may_not_exceed_allowance](#)

§5.13 – No MP accumulation between turns

implemented

manual page unknown

A unit may never accumulate movement points from turn to turn, nor may a unit transfer unused movement points to other units. A unit's unused movement points in any given turn are considered lost.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:2839 GH:omdurman-rules/src/effects.rs:2839	end_player_turn	<pre>2837 /// End the current player's turn: recover disrupted 2838 units, switch active player, advance turn index (rulebook 2839 §4). 2840 pub fn end_player_turn(state: &mut GameState) -> Result<(), RuleError> { 2841 debug!(old_player = ?state.active_player,</pre>

Covered by tests: [omdurman-rules::src::effects::unused_movement_points_do_not_carry_over](#)

§5.21 – Friendlies transport via gunboat

implemented

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In general, naval transport missions are not allowed, i.e. gunboats may not carry any land units. The sole exception is that the Anglo-Egyptian player may transport the surviving units of the “Friendlies” brigade from the east bank of the Nile to the west bank after, and only after, the Dervish east bank unit (Isa Zachneih) has been eliminated. The transport is accomplished in the following sequence: a) on any turn that a “Friendlies” unit and any Anglo-Egyptian gunboat start their turn adjacent, that unit may load onto (i.e. stack with) the gunboat; b) during the Anglo-Egyptian player's next turn the gunboat may move to any Nile hex adjacent to a west bank hex (up to the gunboat's movement

allowance); c) on the Anglo-Egyptian player's third turn the "Friendlies" unit may disembark and move normally, paying the normal terrain cost for the first hex entered. The gunboat may also move normally that turn.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:576 GH:omdurman-rules/src/lib.rs:576	is_friendlylies	<pre> 574 /// "Friendlies" units obey several special rules 575 (\$5.21, \$5.23, \$6.52, 576 /// \$9.14 victory conditions). 577 pub fn is_friendlylies(&self) -> bool { 578 matches!(579 self, </pre>
omdurman-rules/src/effects.rs:4215 GH:omdurman-rules/src/effects.rs:4215	friendlylies_transport_offer	<pre> 4213 /// regardless of selection. Pairs with 4214 [`GameEffect::FriendlyliesTransport`] 4215 /// so the UI can offer exactly the action the 4216 engine would accept. 4217 pub fn friendlylies_transport_offer(&self, selected: 4218 Option<UnitId>) -> Option<FriendlyliesAction> { 4219 match self.friendlylies_transport { 4220 None => { </pre>
omdurman-rules/src/lib.rs:759 GH:omdurman-rules/src/lib.rs:759	loaded_on	<pre> 757 pub disrupted: bool, 758 /// `Some(gunboat)` after a "Friendlies" unit loads 759 onto a gunboat (\$5.21). 760 pub loaded_on: Option<UnitId>, 761 /// Set while the unit is building Zariba hexsides 762 -- neither offensive 763 /// fire nor melee allowed that turn (\$5.3). </pre>
omdurman-rules/src/effects.rs:229 GH:omdurman-rules/src/effects.rs:229	FriendlyliesTransport	<pre> 227 /// Load/disembark the "Friendlies" brigade via 228 gunboat (rulebook \$5.21). 229 FriendlyliesTransport(crate::FriendlyliesAction), 230 231 // -- Optional rules </pre>
omdurman-rules/src/effects.rs:4998 GH:omdurman-rules/src/effects.rs:4998	apply_friendlylies_transport	<pre> 4996 /// unit is freed (a disembarking `MoveUnit` 4997 should follow, costed by 4998 terrain). 4999 pub fn apply_friendlylies_transport(5000 state: &mut GameState, 5001 action: FriendlyliesAction, </pre>
omdurman-rules/src/lib.rs:992 GH:omdurman-rules/src/lib.rs:992	FriendlyliesAction	<pre> 990 /// tracks each unit-gunboat pair independently. 991 #[derive(Serialize, Deserialize, Clone, Copy, 992 PartialEq, Eq, Debug)] 993 pub enum FriendlyliesAction { 994 /// Turn N (the load turn): unit and gunboat 995 started adjacent; unit 996 /// loads onto (stacks with) the gunboat. </pre>
omdurman-rules/src/lib.rs:1012 GH:omdurman-rules/src/lib.rs:1012	TransportState	<pre> 1010 /// third turn. 1011 #[derive(Serialize, Deserialize, Clone, Copy, 1012 PartialEq, Eq, Debug)] 1013 pub enum TransportState { 1014 /// Turn N (the load turn): unit and gunboat 1015 started adjacent; unit 1016 /// loads onto (stacks with) the gunboat. </pre>

Covered by tests: [omdurman-rules::src::effects::friendlylies_transport_offer_load_requires_prerequisites](#)
[omdurman-rules::src::effects::friendlylies_transport_offer_follows_state_machine](#)

§5.22 – Land units may never enter a Nile River hex

[implemented](#)

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With the exception of 5.21, land units may never enter a Nile River hex. Only gunboats may enter and move along Nile River hexes.

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1500 GH:omdurman-rules/src/effects.rs:1500	can_move_unit_to	<pre> 1498 /// 1499 /// [<code>`hex_in_enemy_zoc`</code>]: <code>Self::hex_in_enemy_zoc</code> 1500 pub fn can_move_unit_to(1501 &self, 1502 unit_id: UnitId,</pre>
omdurman-rules/src/effects.rs:1171 GH:omdurman-rules/src/effects.rs:1171	in_deployment_zone	<pre> 1169 /// plan / UI rather than this hex predicate. 1170 Documented, not silently 1171 /// dropped. 1172 pub fn in_deployment_zone(&self, player: Player, 1173 hex: HexCoord, is_boat: bool) -> bool { 1174 // No board attached -> permissive (unit 1175 tests, unbound session). 1176 if self.board.terrain.is_empty() {</pre>

Covered by tests: [omdurman-rules::src::effects::campaign_deployment_is_boat_land_exclusive](#)
[omdurman-rules::src::effects::fok_ae_gunboat_deploys_only_on_nile](#) [omdurman-rules::src::effects::fok_ae_land_unit_rejected_on_nile](#)
[omdurman-rules::src::effects::deploy_via_real_sprite_resolution_matches_engine](#)
[omdurman-rules::src::effects::fok_dervish_land_unit_rejected_on_nile](#)
[omdurman-rules::src::effects::retreat_before_melee_may_not_land_on_nile](#)

§5.23 – Walled city entry restrictions

implemented

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Only certain units may enter the walled portion of Omdurman. For the Dervish player these are the Khalifa unit, the three Dervish artillery units, and the Taiasha units (the Khalifa's bodyguard). Any Anglo-Egyptian units that can get to the walled city may enter it (except gunboats and "Friendlies"). Units entering and/or exiting the walled city may only do so through a gate or breach hexside.

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:130 GH:omdurman-types/src/lib.rs:130	HexsideRef	<pre> 128 /// data by [<code>`HexsideRef`</code>]. 129 #[derive(Serialize, Deserialize, Clone, Copy, 130 PartialEq, Eq, Hash, Debug)] 131 pub struct HexsideRef { 132 pub a: HexCoord,</pre>
omdurman-types/src/lib.rs:165 GH:omdurman-types/src/lib.rs:165	HexsideKind	<pre> 163 strum::EnumIter, 164)] 165 pub enum HexsideKind { 166 /// City wall (Khartoum, walled city of Omdurman). 167 Blocks LOS, blocks 168 /// movement except across gates/breaches (§5.23), 169 blocks ZOC into the city</pre>
omdurman-types/src/lib.rs:236 GH:omdurman-types/src/lib.rs:236	blocks_movement	<pre> 234 /// <code>`omdurman-rules`</code>). The trench <code>*end*</code> variants 235 are therefore intentionally 236 /// not blocking. 237 pub fn blocks_movement(self) -> bool { 238 matches!(239 self,</pre>
omdurman-rules/src/lib.rs:602	may_enter_walled_city	<pre> 600 /// Taiasha bodyguard may enter. Anglo-Egyptian: 601 any unit that can reach the 602 /// walled city <code>*except*</code> gunboats and "Friendlies".</pre>

GH:omdurman-rules/src/lib.rs:602		<pre> 602 pub fn may_enter_walled_city(&self) -> bool { 603 match self { 604 // §5.23 Dervish: Khalifa, artillery, Taiasha. </pre>
omdurman-rules/src/board.rs:294 GH:omdurman-rules/src/board.rs:294	is_walled_city	<pre> 292 /// are its seeds). The set is derived once from the board data, replacing 293 /// the older "at least two of six hexsides are Wall/Gate/Breach" heuristic. 294 pub fn is_walled_city(&self, hex: HexCoord) -> bool { 295 // Membership in the precomputed enclosed area (see `walled_city`). 296 // Palace/Tomb hexes are always part of it (they are the seeds). </pre>
omdurman-rules/src/effects.rs:409 GH:omdurman-rules/src/effects.rs:409	WalledCityEntry	<pre> 407 408 #[error("unit {0:?} is not eligible to enter the walled city of Omdurman at {1:?} (§5.23)")] 409 WalledCityEntry(UnitId, HexCoord), 410 411 #[error("movement cost {cost:?} exceeds allowance {allowance:?}")] </pre>

Covered by tests:

- omdurman-rules::src::effects::can_move_rejects_wall_hexside
- omdurman-rules::src::effects::can_move_allows_gate_hexside
- omdurman-rules::src::effects::walled_city_entry_rejects_unauthorized_dervish
- omdurman-rules::src::effects::walled_city_entry_rejects_ae_gunboat
- omdurman-rules::src::effects::walled_city_entry_not_enforced_for_fok
- omdurman-rules::src::board_data::campaign_walled_city_is_enclosed_by_walls

§5.24 – Gunboat upstream/downstream movement

implemented

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Note that gunboats have two movement allowances separated by a slash, e.g. 10/16. The smaller number is the movement allowance when moving upstream, i.e. against the current (the direction of the current is indicated by arrows in the Nile). The larger number is the movement allowance when moving downstream, i.e. with the current. Gunboats may combine movement in both directions, but if they move even one hex upstream, their upstream movement allowance is their maximum movement allowance for that turn, and may not be exceeded.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:512 GH:omdurman-rules/src/lib.rs:512	GunboatMovement	<pre> 510 /// the turn. 511 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 512 pub struct GunboatMovement { 513 pub upstream: MovementAllowance, 514 pub downstream: MovementAllowance, </pre>
omdurman-types/src/lib.rs:927 GH:omdurman-types/src/lib.rs:927	is_boat	<pre> 925 926 /// Gunboats use the split upstream/downstream movement allowance (§5.24). 927 pub fn is_boat(self) -> bool { 928 matches!(self, UnitKind::Gunboat { .. }) 929 } </pre>

Covered by tests: [omdurman-rules::src::unit_profiles::boat_annotation_yields_split_gunboat_movement](#)
[omdurman-rules::src::effects::gunboat_upstream_cap_is_sticky_across_moves](#)

§5.25 – Dervish forts may not move

[implemented](#)

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Dervish forts may not move in any way once placed.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:743 GH:omdurman-rules/src/lib.rs:743	Immobile	<pre>741 Gunboat(GunboatMovement), 742 /// Forts may not move once placed (§5.25). 743 Immobile, 744 } 745 </pre>
omdurman-types/src/lib.rs:884 GH:omdurman-types/src/lib.rs:884	UnitKind::Fort	<pre>882 /// Permanent emplacement (§6.54): fire (artillery) / melee (defensive). 883 /// May not move once placed (§5.25). 884 Fort { fire: i32, melee: i32 }, 885 /// Dervish leader (§6.51): fire / melee / movement. May melee attack (§7.4). 886 DervishLeader {</pre>
omdurman-rules/src/lib.rs:743 GH:omdurman-rules/src/lib.rs:743	UnitMovement::Immobile	<pre>741 Gunboat(GunboatMovement), 742 /// Forts may not move once placed (§5.25). 743 Immobile, 744 } 745 </pre>

Covered by tests: [omdurman-rules::src::effects::forts_are_never_advance_eligible](#)

§5.26 – Units stop on entering enemy ZOC

[implemented](#)

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Units must stop their movement immediately upon entering an enemy zone of control (see 5.4).

See also: §5.4

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1500 GH:omdurman-rules/src/effects.rs:1500	can_move_unit_to	<pre>1498 /// 1499 /// [<code>hex_in_enemy_zoc</code>]: Self::hex_in_enemy_zoc 1500 pub fn can_move_unit_to(1501 &self, 1502 unit_id: UnitId,</pre>
omdurman-rules/src/effects.rs:2435 GH:omdurman-rules/src/effects.rs:2435	hex_in_enemy_zoc	<pre>2433 /// does not extend into or out of a Nile hex. With no board loaded these 2434 /// reduce to the plain adjacency rule. 2435 pub fn hex_in_enemy_zoc(2436 &self, 2437 hex: HexCoord,</pre>

Covered by tests: [omdurman-rules::src::effects::unit_entering_enemy_zoc_may_move_no_further_that_turn](#)
[omdurman-rules::src::effects::zoc_transit_check_uses_the_actual_path](#)

§5.41 – All units except AE leaders exert ZOC

implemented

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All units except Anglo-Egyptian leaders exert a zone of control (hereafter called a ZOC) into their six adjacent hexes (exception: Gunboats exert a ZOC only against enemy gunboats). Disrupted units have no ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:798 GH:omdurman-rules/src/lib.rs:798	ZocReason	<pre>796 /// Used by the engine when answering "is this hex in 797 an enemy ZOC?". 797 #[derive(Serialize, Deserialize, Clone, Copy, 797 PartialEq, Eq, Debug)] 798 pub enum ZocReason { 799 /// Normal ZOC: any non-disrupted unit other than 800 an Anglo-Egyptian 800 /// leader (§5.41) projects ZOC into each of its 800 six adjacent hexes.</pre>
omdurman-rules/src/effects.rs:2401 GH:omdurman-rules/src/effects.rs:2401	unit_projects_zoc	<pre>2399 /// §5.44) need the game map, which the engine 2400 does not hold; the app layers 2400 /// those on top. This is the position/kind/ 2400 disruption core of the rule. 2401 pub fn unit_projects_zoc(2402 &self, 2403 unit: &UnitPlacement,</pre>
omdurman-rules/src/effects.rs:2476 GH:omdurman-rules/src/effects.rs:2476	zoc_hexes	<pre>2474 /// ZOC covers a given hex; this function returns 2475 *which* hexes a 2475 /// specific unit covers. 2476 pub fn zoc_hexes(2477 &self, 2478 unit: &UnitPlacement,</pre>

Covered by tests: [omdurman-rules::src::effects::zoc_hexes_empty_for_anglo_egyptian_leader](#)
[omdurman-rules::src::effects::zoc_hexes_normal_unit_projects_six_adjacent_minus_exclusions](#)
[omdurman-rules::src::effects::zoc_hexes_empty_for_disrupted_unit](#)
[omdurman-rules::src::effects::zoc_hexes_matches_hex_in_enemy_zoc](#) [omdurman-rules::src::effects::zoc_hexes_excludes_nile](#)
[omdurman-rules::src::effects::zoc_hexes_excludes_khor](#)

§5.42 – No MP cost to enter/leave enemy ZOC

implemented

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There is no movement point cost to enter or leave an enemy ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1666 GH:omdurman-rules/src/effects.rs:1666	movement_cost_for	<pre>1664 /// 1665 /// §5.42: entering or leaving an enemy ZOC adds 1665 no MP cost. 1666 pub fn movement_cost_for(1667 &self, 1668 unit: &UnitPlacement,</pre>

Covered by tests: [omdurman-rules::src::effects::entering_enemy_zoc_costs_no_extra_mp](#)

§5.43 – Units stop when entering enemy ZOC

implemented

manual page unknown

All units must stop when they enter an enemy ZOC and may move no further that turn. In their next movement phase they may withdraw or, if desired, move directly into another enemy ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1500 GH:omdurman-rules/src/effects.rs:1500	can_move_unit_to	<pre>1498 /// 1499 /// [`hex_in_enemy_zoc`]: Self::hex_in_enemy_zoc 1500 pub fn can_move_unit_to(1501 &self, 1502 unit_id: UnitId,</pre>
omdurman-rules/src/effects.rs:2435 GH:omdurman-rules/src/effects.rs:2435	hex_in_enemy_zoc	<pre>2433 /// does not extend into or out of a Nile hex. With no board loaded these 2434 /// reduce to the plain adjacency rule. 2435 pub fn hex_in_enemy_zoc(2436 &self, 2437 hex: HexCoord,</pre>

Covered by tests: [omdurman-rules::src::effects::unit_entering_enemy_zoc_may_move_no_further_that_turn](#)

§5.44 – ZOC limitations (walls, khor, fort, Nile, Zariba)

implemented

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ZOCs do not extend into or out of a Nile River hex (exception: Gunboats, see 5.41). ZOCs do not extend across a khor, into a fort, or into a hex inside the walled city across a wall hexside. ZOCs do extend out of a fort (even if unoccupied), and from a walled city hex into an adjacent non-walled-city hex across a wall hexside. ZOCs also extend out of (but not into) a walled city hex across a gate hexside. ZOCs extend both ways across a breach hexside. ZOCs also extend out of, but not into, a hut or building hex. In the historical scenario ZOCs extend out of, but not into, the Zariba across a Zariba hexside (also in the campaign game if the Zariba is constructed).

(see manual for full text)

See also: §5.41

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:798 GH:omdurman-rules/src/lib.rs:798	ZocReason	<pre>796 /// Used by the engine when answering "is this hex in an enemy ZOC?". 797 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 798 pub enum ZocReason { 799 /// Normal ZOC: any non-disrupted unit other than an Anglo-Egyptian 800 /// leader (§5.41) projects ZOC into each of its six adjacent hexes.</pre>
omdurman-types/src/lib.rs:170 GH:omdurman-types/src/lib.rs:170	Wall	<pre>168 /// (§5.44), blocks melee (§7.2), blocks advance- after-combat (§6.82). 169 #[default] 170 Wall, 171 /// Gate hexside in a wall. ZOC extends *out of* the walled city through 172 /// gates but not into it (§5.44). Melee may be made through a gate (§7.2).</pre>
omdurman-types/src/lib.rs:179	Khor	<pre>177 /// Khor -- gully/wadi. ZOCs do not extend across (§5.44); advance after 178 /// combat may not cross (§6.82).</pre>

GH:omdurman-types/src/lib.rs:179		<pre> 179 Khor, 180 /// Crest line. Blocks LOS unless the firer is on the higher side 181 /// (§6.3 note 7). </pre>
omdurman-rules/src/lib.rs:811 GH:omdurman-rules/src/lib.rs:811	ZocReason::Zariba	<pre> 809 /// Zariba hexside ZOC behaviour in the historical scenario / when the 810 /// Zariba is constructed (§5.44). 811 Zariba, 812 } 813 </pre>
omdurman-rules/src/effects.rs:2401 GH:omdurman-rules/src/effects.rs:2401	unit_projects_zoc	<pre> 2399 /// §5.44) need the game map, which the engine does not hold; the app layers 2400 /// those on top. This is the position/kind/ disruption core of the rule. 2401 pub fn unit_projects_zoc(2402 &self, 2403 unit: &UnitPlacement, </pre>
omdurman-types/src/lib.rs:253 GH:omdurman-types/src/lib.rs:253	HexsideKind::blocks_zoc	<pre> 251 /// cannot express; those are left to the caller. This predicate captures the 252 /// symmetric "does not extend across" cases. 253 pub fn blocks_zoc(self) -> bool { 254 matches!(255 self, </pre>
omdurman-rules/src/effects.rs:2435 GH:omdurman-rules/src/effects.rs:2435	hex_in_enemy_zoc	<pre> 2433 /// does not extend into or out of a Nile hex. With no board loaded these 2434 /// reduce to the plain adjacency rule. 2435 pub fn hex_in_enemy_zoc(2436 &self, 2437 hex: HexCoord, </pre>

Covered by tests: `omdurman-rules::src::effects::zoc_hexes_excludes_nile` `omdurman-rules::src::effects::zoc_hexes_excludes_khor` `omdurman-rules::src::effects::zoc_hexes_normal_unit_projects_six_adjacent_minus_exclusions`

§5.51 – Stacking limit (4 units + leaders, gunboats isolated)

implemented

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No more than four units may occupy a hex, with the exception of leaders and gunboats. All leader units are free stacking, i.e. they may stack in addition to the four-unit-per-hex stacking limitation. Gunboats may not stack with any other unit (Exception: 5.21). Players may move through friendly units at no additional cost in movement points. The stacking limitation applies only at the end of the movement phase and during combat.

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:820 GH:omdurman-rules/src/lib.rs:820	OverLimit	<pre> 818 /// and the gunboat exception. 819 #[error("hex stack exceeds the four-unit limit")] 820 OverLimit, 821 /// "Gunboats may not stack with any other unit" (§5.51, exception §5.21). 822 #[error("gunboats may not stack with non-gunboat units")] </pre>
omdurman-rules/src/lib.rs:823 GH:omdurman-rules/src/lib.rs:823	GunboatStack	<pre> 821 /// "Gunboats may not stack with any other unit" (§5.51, exception §5.21). 822 #[error("gunboats may not stack with non-gunboat units")] 823 GunboatStack, 824 /// "Units of different Dervish tribes may not stack together" (§5.52). 825 #[error("Dervish units of different tribes may not stack")] </pre>

omdurman-rules/src/effects.rs:2230 GH:omdurman-rules/src/effects.rs:2230	check_stacking	<pre>2228 /// * §5.52 -- units of different Dervish tribes may not stack together. 2229 /// * §5.53 -- a Dervish leader may stack only with units of its command. 2230 pub fn check_stacking(2231 &self, 2232 mover: &UnitPlacement,</pre>
---	-----------------------	---

Covered by tests:

- [omdurman-rules::src::effects::stacking_over_limit_rejected](#)
- [omdurman-rules::src::effects::mid_move_stacking_allows_pass_through](#)
- [omdurman-rules::src::effects::mid_move_stacking_rejects_over_limit_destination](#)
- [omdurman-rules::src::effects::validate_stacking_invariants_clean_state](#)
- [omdurman-rules::src::effects::validate_stacking_invariants_catches_stacking_violation](#)
- [omdurman-rules::src::effects::validate_stacking_invariants_allows_leaders_stacking](#)

§5.52 – Different Dervish tribes may not stack together

implemented

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The units of different Dervish tribes may not stack together, even if they are the same color (e.g. although both are green, Mulazmin and Jehadia units may not stack with each other).

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:826 GH:omdurman-rules/src/lib.rs:826	DervishTribeMix	<pre>824 /// "Units of different Dervish tribes may not stack together" (§5.52). 825 #[error("Dervish units of different tribes may not stack")] 826 DervishTribeMix, 827 /// "If Dervish leaders elect to stack, they may only stack with units of 828 /// their command (i.e. colour)" (§5.53).</pre>

Covered by tests:

- [omdurman-rules::src::unit_profiles::green_sections_are_mulazmin_tribal_units](#)
- [omdurman-rules::src::effects::deploy_rejects_dervish_tribe_mix](#)
- [omdurman-rules::src::effects::validate_stacking_invariants_clean_state](#)

§5.53 – Leader stacking with command colour only

implemented

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Leader units are not required to stack. If Dervish leaders elect to stack, however, they may only stack with units of their command (i.e. color). For example, Sheik El Din may only stack with Mulazmins or Jehadias.

****5.54) Anglo-Egyptian Brigade Integrity:**** All British, Sudanese, and Egyptian infantry units have their brigade designation printed in the upper right corner (e.g. “2B” = 2nd British Brigade; “3E” = 3rd Egyptian Brigade, etc.). In any combat phase in which all four infantry battalions belonging to any Anglo-Egyptian infantry brigade are stacked in the same hex they are said to have brigade integrity. Stacks having brigade integrity receive a +1 modifier to their fire combat die roll provided they all fire at the same enemy occupied hex. This modifier is in addition to the normal +1 bonus given to all Anglo-Egyptian direct fire attacks (see 6.24).

(see manual for full text)

See also: §5.54, §6.24

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:830	DervishLeaderCommandMismatch	<pre>828 /// their command (i.e. colour)" (§5.53). 829 #[error("Dervish leader may only stack with units</pre>

GH:omdurman-rules/src/lib.rs:830		<pre> of their own command")) 830 DervishLeaderCommandMismatch, 831 } 832 </pre>
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Covered by tests: `omdurman-rules::src::effects::dervish_leader_stacks_only_with_command_colour`

§5.54 – Anglo-Egyptian Brigade Integrity

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:836 GH:omdurman-rules/src/lib.rs:836	BrigadeIntegrity	<pre> 834 /// stack contains all four battalions of a single Anglo-Egyptian brigade. 835 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 836 pub enum BrigadeIntegrity { 837 None, 838 Integrated(BrigadeId), </pre>
omdurman-rules/src/lib.rs:701 GH:omdurman-rules/src/lib.rs:701	brigade_integrity	<pre> 699 /// Only a full stack of four battalions qualifies. Three or fewer may still 700 /// stack and fire, but they receive no brigade- integrity bonus. 701 pub fn brigade_integrity(identities: &[UnitIdentity]) - > BrigadeIntegrity { 702 let Some(brigade) = identities.first().and_then(i i.brigade()) else { 703 return BrigadeIntegrity::None; </pre>
omdurman-rules/src/lib.rs:836 GH:omdurman-rules/src/lib.rs:836	FireModifier::BrigadeIntegrity	<pre> 834 /// stack contains all four battalions of a single Anglo-Egyptian brigade. 835 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 836 pub enum BrigadeIntegrity { 837 None, 838 Integrated(BrigadeId), </pre>
omdurman-rules/src/lib.rs:315 GH:omdurman-rules/src/lib.rs:315	BattalionOrdinal	<pre> 313 /// brigade integrity requires all four stacked in one hex (§5.54). 314 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 315 pub enum BattalionOrdinal { 316 First = 1, 317 Second = 2, </pre>
omdurman-types/src/lib.rs:992 GH:omdurman-types/src/lib.rs:992	BrigadeId	<pre> 990 /// same field for uniform handling. 991 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 992 pub struct BrigadeId { 993 pub number: u8, 994 pub nationality: BrigadeNationality, </pre>

Covered by tests:

- `omdurman-rules::src::unit_profiles::brigade_designation_ignored_for_non_infantry`
- `omdurman-rules::src::unit_profiles::printed_brigade_designation_overrides_column`
- `omdurman-rules::src::unit_profiles::tribe_stats_come_from_annotation`
- `omdurman-rules::src::unit_profiles::section_owner_anglo_egyptian_sections`
- `omdurman-rules::src::unit_profiles::section_owner_green_sections_are_dervish`
- `omdurman-rules::src::lib::brigade_integrity_four_battalions_returns_integrated`
- `omdurman-rules::src::unit_profiles::ae_infantry_fourth_battalion_from_col_3`
- `omdurman-rules::src::lib::brigade_integrity_empty_slice`
- `omdurman-rules::src::lib::brigade_integrity_friendlyes_returns_none`
- `omdurman-rules::src::unit_profiles::section_owner_dervish_sections`
- `omdurman-rules::src::lib::brigade_integrity_three_battalions_returns_none`
- `omdurman-rules::src::lib::unit_identity_brigade_and_battalion_accessors`
- `omdurman-rules::src::unit_profiles::ae_infantry_brigade_number_three_from_col_7`
- `omdurman-rules::src::lib::brigade_integrity_non_infantry_returns_none`

omdurman-rules::src::lib::brigade_integrity_mixed_brigades_returns_none
omdurman-rules::src::unit_profiles::ae_infantry_third_battalion_from_col_2
omdurman-rules::src::effects::brigade_integrity_modifier_is_engine_derived

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§6 – Fire Combat Phase

§6 – Fire Combat Phase

descriptive

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Fire Combat Phase

§6.1 – General Rules

descriptive

manual page unknown

General Rules

§6.2 – How To Have Fire Combat

descriptive

manual page unknown

How To Have Fire Combat

§6.3 – Line of Sight Table

implemented

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Line of Sight Table

This table is located on the back of this rulebook and should be self-explanatory. Locate the terrain type the firing unit is in and cross-index it with the terrain type the target unit is in. Terrain types in the intersecting box block line of sight, with exceptions as footnoted. Also study the “Special LOS Notes” given and remember that (with the exception of howitzer fire — see 6.64) you can’t fire at anything you can’t see!

See also: §6.64

File	Symbol	Code Snippet
omdurman-rules/src/los_table.rs:57 GH:omdurman-rules/src/los_table.rs:57	LosLevel	<pre>55 serde::Serialize, serde::Deserialize, Clone, Copy, 56 PartialEq, Eq, PartialOrd, Ord, Hash, Debug, 57)] 58 pub enum LosLevel { 59 Ground, 59 Rough,</pre>
omdurman-rules/src/los_table.rs:69 GH:omdurman-rules/src/los_table.rs:69	LosFeature	<pre>67 /// authored RON spellings. 68 #[derive(serde::Serialize, serde::Deserialize, Clone, 69 Copy, PartialEq, Eq, Hash, Debug)] 70 pub enum LosFeature { 71 /// A hex containing units (gunboats/forts excluded 71 Units,</pre>
omdurman-rules/src/los_table.rs:91 GH:omdurman-rules/src/los_table.rs:91	LosCondition	<pre>89 /// A positional condition from the LOS table Detail 90 footnotes. 91 #[derive(serde::Serialize, serde::Deserialize, Clone, 92 Copy, PartialEq, Eq, Hash, Debug)] 93 pub enum LosCondition { 94 /// (1) Blocks only if the ray passes through more</pre>

		than two such features. <pre>93 MoreThanTwo,</pre>
omdurman-rules/src/los_table.rs:170 GH:omdurman-rules/src/los_table.rs:170	los_level	<pre>168 /// 169 /// For all other units, the level is derived from the terrain at `hex`. 170 pub fn los_level_for_unit(171 kind: UnitKind, 172 hex: HexCoord,</pre>
omdurman-rules/src/los_table.rs:170 GH:omdurman-rules/src/los_table.rs:170	los_level_for_unit	<pre>168 /// 169 /// For all other units, the level is derived from the terrain at `hex`. 170 pub fn los_level_for_unit(171 kind: UnitKind, 172 hex: HexCoord,</pre>
omdurman-rules/src/los_table.rs:209 GH:omdurman-rules/src/los_table.rs:209	blocking_rules	<pre>207 /// of its conditions are satisfied (AND semantics); an empty conditions list 208 /// means the feature always blocks. 209 pub fn blocking_rules(firer: LosLevel, target: LosLevel) -> &'static [BlockingRule] { 210 let table = crate::tables_data::los_table_data(); 211 match table.cells.get(&(firer, target)) {</pre>
omdurman-rules/src/los_table.rs:295 GH:omdurman-rules/src/los_table.rs:295	has_los	<pre>293 /// `unit_level_at` closure returns the LOS level of blocking units 294 /// (non-gunboat, non-fort per note a) in an intervening hex, or `None`. 295 pub fn has_los(296 board: &crate::board::BoardInfo, 297 from: HexCoord,</pre>
omdurman-types/src/lib.rs:208 GH:omdurman-types/src/lib.rs:208	HexsideKind::blocks_los	<pre>206 /// (LOS table conditions 2-4, 7) and note (e) are handled by the engine 207 /// in `omdurman_rules::los_table`, not by this predicate. 208 pub fn blocks_los(self) -> bool { 209 matches!(self, HexsideKind::Wall HexsideKind::Crest) 210 }</pre>
omdurman-types/src/lib.rs:208 GH:omdurman-types/src/lib.rs:208	Terrain::blocks_los	<pre>206 /// (LOS table conditions 2-4, 7) and note (e) are handled by the engine 207 /// in `omdurman_rules::los_table`, not by this predicate. 208 pub fn blocks_los(self) -> bool { 209 matches!(self, HexsideKind::Wall HexsideKind::Crest) 210 }</pre>
omdurman-types/src/lib.rs:430 GH:omdurman-types/src/lib.rs:430	Terrain::is_los_trees	<pre>428 /// (\$6.3 note 1). Retained for compatibility; the full LOS engine 429 /// checks `Terrain::Trees` directly. 430 pub fn is_los_trees(self) -> bool { 431 matches!(self, Terrain::Trees { .. }) 432 }</pre>

Covered by tests: [omdurman-rules::src::los_table::los_level_mapping](#) [omdurman-rules::src::los_table::blocking_rules_all_cells_covered](#) [omdurman-rules::src::los_table::has_los_empty_board_is_clear](#) [omdurman-rules::src::los_table::has_los_adjacent_clear](#) [omdurman-rules::src::los_table::has_los_howitzzer_bypasses](#) [omdurman-rules::src::los_table::has_los_wall_hexside_blocks_ground_to_ground](#) [omdurman-rules::src::los_table::has_los_gate_hexside_passes](#) [omdurman-rules::src::los_table::has_los_breach_hexside_passes](#) [omdurman-rules::src::los_table::has_los_rough_intervening_blocks_ground_to_ground](#) [omdurman-rules::src::los_table::has_los_two_tree_hexes_pass_ground_to_ground](#) [omdurman-rules::src::los_table::has_los_three_tree_hexes_block_ground_to_ground](#) [omdurman-rules::src::los_table::has_los_two_hut_hexes_pass_ground_to_ground](#) [omdurman-rules::src::los_table::has_los_three_hut_hexes_block_ground_to_ground](#) [omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_clear_no_units](#) [omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_blocked_by_hilltop_unit](#)

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omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_not_blocked_by_ground_unit
omdurman-rules::src::los_table::has_los_rough_to_rough_unit_at_lower_level_passes
omdurman-rules::src::los_table::has_los_rough_to_rough_unit_at_same_level_blocks
omdurman-rules::src::los_table::has_los_rough_to_rough_hilltop_blocks
omdurman-rules::src::los_table::has_los_ground_to_hilltop_intervening_hilltop_blocks
omdurman-rules::src::los_table::has_los_building_blocks_like_huts_ground_to_ground
omdurman-rules::src::los_table::has_los_two_building_hexes_pass_ground_to_ground
omdurman-rules::src::los_table::los_level_for_unit_gunboat_is_rough
omdurman-rules::src::los_table::los_level_for_unit_fort_is_ground
omdurman-rules::src::los_table::los_level_for_unit_walled_city_adj_wall_is_rough
omdurman-rules::src::los_table::gunboat_firer_uses_rough_row_not_ground
omdurman-rules::src::los_table::los_reflexive_all_terrains
omdurman-rules::src::los_table::los_reflexive_hilltop
omdurman-rules::src::los_table::los_symmetric_ground_to_ground_no_units
omdurman-rules::src::los_table::los_howitzer_always_has_los
omdurman-rules::src::los_table::los_howitzer_same_hex
omdurman-rules::src::los_table::los_blocking_rules_match_reference_table
omdurman-rules::src::los_table::los_ground_to_ground_features_block_as_expected

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§6.4 – Fire Combat Sequence

implemented

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Fire Combat Sequence

The sequence of fire combat resolution is the same for both defensive and offensive fire combat. During the Dervish player turn the Anglo-Egyptian player executes 6.41 AND 6.42 as defensive fire, after which the Dervish player executes 6.41 as offensive fire. During the Anglo-Egyptian player turn the Dervish player executes 6.41 as defensive fire, after which the Anglo-Egyptian player executes 6.41 AND 6.42 as offensive fire.

****6.41) Direct Fire Subphase (Dervish and Anglo-Egyptian players):**** The firing player must first allocate all of his fire attacks, combining his units' direct fire combat factors in any manner he wishes. After all fire has been allocated, the firing player then resolves his attacks in any order he wishes.

****6.42) Maxim Second Fire and Howitzer Fire Subphase (Anglo-Egyptian player only):**** Anglo-Egyptian named gunboats may now fire their artillery factor as howitzer fire (see 6.64) and all Maxim guns may fire a second time. Once

(see manual for full text)

See also: §6.41, §6.42, §6.64

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:288 GH:omdurman-rules/src/lib.rs:288	FireSubPhase	<pre> 286 287 #[derive(Serialize, Deserialize, Clone, Copy, 288 PartialEq, Eq, Debug)] 289 pub enum FireSubPhase { 290 /// Direct fire (§6.41). Both sides participate in this sub-phase. 291 DirectFire, </pre>
omdurman-rules/src/effects.rs:105 GH:omdurman-rules/src/effects.rs:105	FireCombat	<pre> 103 /// - Firers marked as fired; target hex marked as fired-at. 104 /// - Victory points awarded for eliminations. 105 FireCombat { attack: FireAttack, roll: DieRoll }, 106 107 /// Resolve a howitzer bombardment (two rolls: CRT + impact scatter) </pre>

Covered by tests: [omdurman-rules::src::effects::turn_advances_through_phases](#)

§6.5 – Special Unit Capabilities

descriptive

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Special Unit Capabilities

****6.51) Leader Units:**** Dervish leader units have a fire factor, a melee factor, and a movement factor. They may thus attack, melee, and be eliminated like any other combat unit. Their special benefit is that they stack free.

Anglo-Egyptian leaders have a movement factor only. They are eliminated if a) they are alone in a hex when a Dervish unit occupies or passes through that hex, or b) if all of the combat units a leader is stacked with are eliminated in fire combat or melee. The special function of Anglo-Egyptian leaders is that at least one must survive to occupy the Mahdi's tomb hex if it is to be taken from the Dervish player (see 9.14).

****6.52) Anglo-Egyptian "Friendlies" Brigade:**** These units represent native volunteers in the Anglo-Egyptian army. They fire rifles on the Dervish Range Effects Table and melee with the Dervish melee modifier. They may not enter the walled city of Omdurman (see 5.23). They may be transferred to the west bank (see 5.21).

(see manual for full text)

See also: §5.21, §5.23, §5.44, §6.51, §6.52, §6.53, §6.54, §6.62, §6.63, §7.6, §9.14

§6.6 – Special Artillery Capabilities

descriptive
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Special Artillery Capabilities

§6.7 – Defensive Fire

implemented
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Defensive Fire

In Defensive Fire phase, all of the non-moving player's units may fire at any of the moving player's units in range, within the limitations imposed by the rules of combat (see 6.1 to 6.6). There is no advance after combat as a result of defensive fires.

See also: §6.1, §6.6

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:4084 GH:omdurman-rules/src/effects.rs:4084	can_advance_after_combat	4082 // player's unit, not artillery, adjacent to 4083 `to`, and `to` now empty. 4084 // Wall/khor hexside restrictions are not 4085 enforced (no hexside map data). 4086 pub fn can_advance_after_combat(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 4087 let unit = self.unit_or_err(unit_id)?; 4088 // §6.7: there is no advance after combat as a 4089 result of defensive fire.

Covered by tests: omdurman-rules::src::effects::no_advance_after_defensive_fire
omdurman-rules::src::effects::defensive_fire_opens_no_advance_window

§6.8 – Offensive Fire

descriptive
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Offensive Fire

§6.11 – Fire combat factor printed on units

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The fire combat factor of the various unit types is printed directly on the units and is a numerical expression of the unit's fire strength.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:80 GH:omdurman-rules/src/lib.rs:80	FireFactor	<pre>78 /// Every possible value from the annotated counter 79 set is a named variant. 80 #[derive(Serialize, Deserialize, Clone, Copy, 81 PartialEq, Eq, Hash, Debug, strum::Display)] 82 pub enum FireFactor { 83 One = 1, 84 Three = 3,</pre>

Covered by tests: [omdurman-rules::src::lib::fire_factor_sum_to_row](#)

§6.12 – Fire combat is always voluntary

implicit

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Fire combat is always voluntary. A unit is never required to fire at enemy units merely because they are in range or adjacent.

§6.13 – Fire factor is unitary (may not be divided)

implicit

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If a unit elects to fire, its fire combat factor at an enemy unit, that fire combat factor is unitary. A unit's fire combat factor may not be divided up to fire at enemy units on different hexes.

§6.14 – Players may combine fire factors into one attack

implemented

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Players may combine fire during fire combat phase, i.e. they may fire at an enemy-occupied hex with as many friendly units as may legally do so, combining all of their fire combat factors into one attack. Note that in any given fire combat phase, however, a combat unit may only fire once and may only be fired at once (exceptions: Maxim guns and gunboats — see 6.4).

See also: §6.4

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:94 GH:omdurman-rules/src/lib.rs:94	sum_to_row	<pre>92 impl FireFactor { 93 /// Sum multiple fire factors and return the 94 corresponding Combat Results Table row (rulebook §6.11). 95 pub fn sum_to_row<'a>(factors: impl 96 IntoIterator<Item = &'a FireFactor>) -> FireFactorRow { 97 let total: u16 = factors.into_iter().map(f 98 f.value()).sum(); 99 crate::combat_results_table::FireFactorRow::from_total(total)</pre>

Covered by tests: [omdurman-rules::src::effects::unit_may_only_be_fired_at_once_per_phase](#)
[omdurman-rules::src::effects::gunboat_and_maxim_may_be_fired_at_repeatedly](#)

§6.15 – May divide a stack to fire at different hexes

implicit

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Players may also divide a stack of units in order to fire at different enemy-occupied hexes. Anglo-Egyptian infantry units having brigade integrity, however, do not receive their +1 direct fire modifier unless they all fire at the same enemy-occupied hex (see 5.54).

See also: §5.54

§6.16 – Halving fire strength rounds down, minimum 1

implemented

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When halving fire combat strength, always round down each individual unit. For example, an Egyptian brigade of four battalions, each having a printed strength of 9 fire factors, will fire a total of 16 factors when halved. However, a unit's firing strength is never reduced below one by halving.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:479 GH:omdurman-rules/src/lib.rs:479	RangeBand	<pre>477 /// multiplied at a given distance (§6.22). 478 #[derive(Serialize, Deserialize, Clone, Copy, 479 pub enum RangeBand { 480 Tripled, 481 Doubled,</pre>

Covered by tests: [omdurman-rules::src::lib::halving_rounds_down_and_never_below_one](#)

§6.21 – First check LOS before firing

implemented

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When combat units wish to fire at enemy units, first check the Line of Sight Table to be sure the firing unit can see the target hex (exception: howitzer fire, see 6.64).

See also: §6.64

File	Symbol	Code Snippet
omdurman-rules/src/los_table.rs:209 GH:omdurman-rules/src/los_table.rs:209	blocking_rules	<pre>207 /// of its conditions are satisfied (AND semantics); an empty conditions list 208 /// means the feature always blocks. 209 pub fn blocking_rules(firer: LosLevel, target: LosLevel) -> &'static [BlockingRule] { 210 let table = crate::tables_data::los_table_data(); 211 match table.cells.get(&(firer, target)) {</pre>
omdurman-rules/src/los_table.rs:295 GH:omdurman-rules/src/los_table.rs:295	has_los	<pre>293 /// `unit_level_at` closure returns the LOS level of blocking units 294 /// (non-gunboat, non-fort per note a) in an intervening hex, or `None`. 295 pub fn has_los(296 board: &crate::board::BoardInfo, 297 from: HexCoord,</pre>

Covered

by

tests:

[omdurman-rules::src::effects::can_fire_at_rejects_blocked_los](#) [omdurman-rules::src::effects::can_fire_at_allows_clear_los](#)

§6.22 – Consult Range Effects Table

implemented

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Next consult the Range Effects Table to see if the firing unit's fire combat factor is tripled, doubled, normal, halved, or if the target hex is out of range. Add up the total number of fire combat factors firing at the enemy-occupied hex.

File	Symbol	Code Snippet
omdurman-rules/src/range_effects.rs:35 GH:omdurman-rules/src/range_effects.rs:35	ae_range_effects	<pre>33 /// Look up the range band for an Anglo-Egyptian weapon (§6.22, §6.24). 34 /// Distances > 10 are out of range for all weapons. 35 pub fn ae_range_effects(weapon: WeaponClass, distance: HexDistance) -> RangeBand { 36 band_at(faction_rows(true), weapon, distance) 37 }</pre>
omdurman-rules/src/range_effects.rs:41 GH:omdurman-rules/src/range_effects.rs:41	dervish_range_effects	<pre>39 /// Look up the range band for a Dervish weapon (§6.22). 40 /// Distances > 10 are out of range for all weapons. 41 pub fn dervish_range_effects(weapon: WeaponClass, distance: HexDistance) -> RangeBand { 42 band_at(faction_rows(false), weapon, distance) 43 }</pre>
omdurman-rules/src/lib.rs:479 GH:omdurman-rules/src/lib.rs:479	RangeBand	<pre>477 /// multiplied at a given distance (§6.22). 478 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 479 pub enum RangeBand { 480 Tripled, 481 Doubled,</pre>
omdurman-rules/src/lib.rs:181 GH:omdurman-rules/src/lib.rs:181	HexDistance	<pre>179 /// (rulebook §6.22, §7.5). 180 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 181 pub struct HexDistance(u16); 182 183 impl HexDistance {</pre>

Covered by tests:

[omdurman-rules::src::range_effects::ae_rifles_doubled_at_range_1](#) [omdurman-rules::src::range_effects::ae_rifles_halved_at_range_4](#)

[omdurman-rules::src::range_effects::ae_howitzer_range](#) [omdurman-rules::src::range_effects::dervish_rifles_shorter_range](#)

[omdurman-rules::src::range_effects::melee_only_range_1](#) [omdurman-rules::src::range_effects::ae_range_effects_artillery_full](#)

[omdurman-rules::src::range_effects::ae_range_effects_maxims_match_rifles](#)

[omdurman-rules::src::range_effects::ae_range_effects_distance_over_10](#)

[omdurman-rules::src::range_effects::ae_range_effects_howitzer_halved_4_to_10](#)

[omdurman-rules::src::range_effects::dervish_range_effects_rifles](#) [omdurman-rules::src::range_effects::dervish_range_effects_artillery](#)

[omdurman-rules::src::range_effects::dervish_range_effects_maxims_and_howitzer](#)

[omdurman-rules::src::range_effects::dervish_range_effects_melee](#)

[omdurman-rules::src::range_effects::dervish_range_effects_distance_over_10](#)

[omdurman-rules::src::effects::fire_combat_eliminate_target](#) [omdurman-rules::src::range_effects::max_day_range_all_combos](#)

[omdurman-rules::src::range_effects::range_effects_every_cell_ae_artillery](#)

[omdurman-rules::src::range_effects::range_effects_every_cell_ae_howitzer](#)

[omdurman-rules::src::range_effects::range_effects_every_cell_ae_maxims](#)

[omdurman-rules::src::range_effects::range_effects_every_cell_ae_rifles](#)

[omdurman-rules::src::range_effects::range_effects_every_cell_dervish_artillery](#)

[omdurman-rules::src::range_effects::range_effects_every_cell_dervish_maxims_howitzer](#)

[omdurman-rules::src::range_effects::range_effects_every_cell_dervish_rifles](#)

[omdurman-rules::src::range_effects::range_effects_every_cell_dervish_spears](#)

[omdurman-rules::src::effects::can_fire_at_gates_phase_range_and_player](#) [omdurman-rules::src::effects::mixed_attack_bands_per_firer](#)

[omdurman-rules::src::range_effects::ae_range_effects_monotone_non_increasing](#)

[omdurman-rules::src::range_effects::ae_howitzer_has_minimum_range](#)

[omdurman-rules::src::range_effects::dervish_range_effects_monotone_non_increasing](#)

[omdurman-rules::src::range_effects::range_effects_first_range_max_effect_last_range_oor](#)

§6.23 – Terrain defensive modifier

implemented

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Next check the Terrain Effects Chart to see if the enemy-occupied hex fired upon contains any terrain which gives the enemy units in that hex a defensive benefit. If so, apply this negative modifier to the roll of the ten-sided die and cross-index your net die roll on the Combat Results Table with the number of combat factors firing.

File	Symbol	Code Snippet
omdurman-rules/src/terrain_chart.rs:59 GH:omdurman-rules/src/terrain_chart.rs:59	defense_modifier	<pre>57 58 /// Convenience: get the defense modifier for a terrain type (rulebook §6.23, Terrain Effects Chart). 59 pub fn defense_modifier(terrain: Terrain) -> i16 { 60 terrain_effects_chart(terrain).defense_modifier 61 }</pre>
omdurman-rules/src/lib.rs:857 GH:omdurman-rules/src/lib.rs:857	FireModifier::Terrain	<pre>855 /// Negative modifier from the Terrain Effects Chart applied to the 856 /// defender's hex (§6.23). 857 Terrain(i16), 858 /// -2 thorn-hedge defensive modifier (§9.231). 859 ZaribaThornHedge,</pre>

Covered

by tests: [omdurman-rules::src::terrain_chart::clear_terrain_no_bonus](#) [omdurman-rules::src::terrain_chart::building_gives_minus_3](#)
[omdurman-rules::src::terrain_chart::palm_grove_gives_minus_2](#) [omdurman-rules::src::terrain_chart::rough_movement_and_defense](#)
[omdurman-rules::src::terrain_chart::swamp_movement_and_defense](#)
[omdurman-rules::src::terrain_chart::hilltop_movement_and_defense](#)
[omdurman-rules::src::terrain_chart::huts_movement_and_defense](#)
[omdurman-rules::src::terrain_chart::defense_modifier_convenience_matches_chart](#)
[omdurman-rules::src::terrain_chart::terrain_movement_costs_in_bounds](#)
[omdurman-rules::src::terrain_chart::terrain_defense_modifier_non_positive](#)

§6.24 – Anglo-Egyptian direct fire accuracy bonus and brigade integrity

implemented

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All Anglo-Egyptian direct fire attacks receive a +1 modifier to their die roll as an accuracy bonus. In addition, any stack of Anglo-Egyptian infantry having brigade integrity (see 5.54) receives a +1 modifier to their die roll if all four fire at the same enemy-occupied hex. These modifiers are cumulative.

See also: §5.54

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:851 GH:omdurman-rules/src/lib.rs:851	AngloEgyptianDirectFire	<pre>849 pub enum FireModifier { 850 /// +1 to all Anglo-Egyptian *direct* fire (§6.24). 851 AngloEgyptianDirectFire, 852 /// +1 brigade integrity, applied only if all four battalions fire at 853 /// the same enemy-occupied hex (§5.54, §6.24).</pre>
omdurman-rules/src/lib.rs:836 GH:omdurman-rules/src/lib.rs:836	BrigadeIntegrity	<pre>834 /// stack contains all four battalions of a single Anglo-Egyptian brigade. 835 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 836 pub enum BrigadeIntegrity { 837 None, 838 Integrated(BrigadeId),</pre>
omdurman-rules/src/lib.rs:946	FireModifier::die_modifier	<pre>944 impl MeleeModifier { 945 /// Return the numeric die-roll modifier for this melee bonus/penalty (rulebook §7.7, §9.232).</pre>

GH:omdurman-rules/src/lib.rs:946		<pre> 946 pub fn die_modifier(self) -> i16 { 947 match self { 948 MeleeModifier::DervishStandard => 2, </pre>
----------------------------------	--	---

Covered by tests: `omdurman-rules::src::effects::brigade_integrity_modifier_is_engine_derived`
`omdurman-rules::src::effects::fire_modifiers_are_engine_derived_and_mismatches_rejected`

§6.41 – Direct Fire Subphase

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:290 GH:omdurman-rules/src/lib.rs:290	DirectFire	<pre> 288 pub enum FireSubPhase { 289 /// Direct fire (§6.41). Both sides participate in 290 this sub-phase. 291 DirectFire, 292 /// Anglo-Egyptian only: Maxim second fire + named- 293 gunboat howitzer fire (§6.42). 294 MaximSecondAndHowitzer, </pre>

Covered by tests: `omdurman-rules::src::effects::fire_combat_elimimates_target`

§6.42 – Maxim Second Fire and Howitzer Fire Subphase

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:292 GH:omdurman-rules/src/lib.rs:292	MaximSecondAndHowitzer	<pre> 290 DirectFire, 291 /// Anglo-Egyptian only: Maxim second fire + named- 292 gunboat howitzer fire (§6.42). 293 MaximSecondAndHowitzer, 294 } </pre>
omdurman-types/src/lib.rs:945 GH:omdurman-types/src/lib.rs:945	fires_twice	<pre> 943 /// Maxim guns fire twice per turn -- once in the 944 Direct Fire Subphase and 945 /// again in the Maxim Second Fire Subphase 946 (rulebook §6.42). 947 pub fn fires_twice(self) -> bool { 948 matches!(self, UnitKind::Maxim { .. }) 949 } </pre>

Covered by tests: `omdurman-rules::src::howitzer_scatter::howitzer_on_target_7_to_10`
`omdurman-rules::src::howitzer_scatter::howitzer_scatters_below_7`
`omdurman-rules::src::effects::advance_window_bridges_fire_subphase_and_closes_at_melee`
`omdurman-rules::src::effects::fired_at_tracker_resets_at_maxim_subphase`

§6.51 – Leader Units

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:403 GH:omdurman-rules/src/lib.rs:403	BritishLeader	<pre> 401 /// to claim the Mahdi's Tomb (§9.14). 402 #[derive(Serialize, Deserialize, Clone, Copy, 403 PartialEq, Eq, Hash, Debug, strum::Display)] 404 pub enum BritishLeader { 405 Kitchener, 406 Gatacre, </pre>

omdurman-types/src/lib.rs:892 GH:omdurman-types/src/lib.rs:892	BritishLeader	<pre> 890 }, 891 /// Anglo-Egyptian leader (§6.51): movement only. 892 BritishLeader { movement: i32 }, 893 /// Wall-breach marker placed by artillery fire 894 Breech,</pre>
omdurman-types/src/lib.rs:933 GH:omdurman-types/src/lib.rs:933	has_combat_factors	<pre> 931 /// British leaders print a movement factor only 932 (§6.51); other kinds carry 933 /// fire and/or melee factors. Markers carry no 934 stats. 935 pub fn has_combat_factors(self) -> bool { 936 !matches!(937 self,</pre>

Covered by tests: [omdurman-rules::src::unit_profiles::zero_factor_is_none_not_zero](#)
[omdurman-rules::src::unit_profiles::kitchener_block_resolves_leaders_friendlys_camel_and_sudanese](#)
[omdurman-rules::src::unit_profiles::dervish_leader_sections_resolve_leader_and_retinue_per_cell](#)

§6.52 – Anglo-Egyptian Friendlys Brigade

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:576 GH:omdurman-rules/src/lib.rs:576	is_friendlys	<pre> 574 /// "Friendlys" units obey several special rules 575 (§5.21, §5.23, §6.52, 576 /// §9.14 victory conditions). 577 pub fn is_friendlys(&self) -> bool { 578 matches!(579 self,</pre>
omdurman-types/src/lib.rs:964 GH:omdurman-types/src/lib.rs:964	Friendlys	<pre> 962 /// Native volunteer brigade -- the Shaggyeh 963 (§6.52). Do not receive 964 /// brigade integrity (§5.54 enumerates only 965 British/Egyptian/Sudanese). 966 Friendlys, 967 }</pre>

Covered by tests: [omdurman-rules::src::unit_profiles::friendlys_counters_score_by_bank_not_as_leaders](#)
[omdurman-rules::src::effects::friendlys_validate_and_resolve_on_dervish_table](#)

§6.53 – Royal Engineers demolition

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:547 GH:omdurman-rules/src/lib.rs:547	RoyalEngineers	<pre> 545 /// The Royal Engineers (§6.53) -- a *specific* 546 unit, not a class, so we 547 /// model it explicitly. 548 RoyalEngineers, 549 }</pre>
omdurman-rules/src/lib.rs:765 GH:omdurman-rules/src/lib.rs:765	demolishing	<pre> 763 /// Set when the Royal Engineers are committed to a 764 demolition this turn 765 /// (§6.53) -- neither offensive fire nor melee 766 allowed that turn. 767 pub demolishing: bool, 768 /// Set when a gunboat has lost its engines to a 769 river mine (§10.12, roll 770 5-7): it may no longer move under power and 771 instead drifts two hexes per</pre>

omdurman-rules/src/effects.rs:208 GH:omdurman-rules/src/effects.rs:208	Demolition	<pre> 206 207 /// Royal Engineers demolition (rulebook §6.53). 208 Demolition { 209 unit_id: UnitId, 210 target: DemolitionTarget, </pre>
omdurman-rules/src/effects.rs:4590 GH:omdurman-rules/src/effects.rs:4590	apply_demolition	<pre> 4588 /// resolution happens at end of turn via 4589 [`apply_resolve_demolition`], which 4590 /// checks the engineer is still adjacent and 4591 undisrupted. 4592 pub fn apply_demolition(4593 state: &mut GameState, 4594 unit_id: UnitId, </pre>
omdurman-rules/src/lib.rs:976 GH:omdurman-rules/src/lib.rs:976	DemolitionTarget	<pre> 974 /// disrupted or driven off. 975 #[derive(Serialize, Deserialize, Clone, Copy, 976 PartialEq, Eq, Debug)] 977 pub enum DemolitionTarget { 978 Fort(UnitId), 979 WallHexside(HexsideRef), </pre>
omdurman-rules/src/effects.rs:4180 GH:omdurman-rules/src/effects.rs:4180	demolition_targets	<pre> 4178 /// the rules would accept. Empty when the unit 4179 doesn't exist or has no 4180 adjacent target. 4181 pub fn demolition_targets(&self, unit_id: UnitId) 4182 -> Vec<DemolitionTarget> { 4183 let Ok(unit) = self.unit_or_err(unit_id) else 4184 { 4185 return Vec::new(); </pre>

Covered by tests: [omdurman-rules::src::effects::demolition_cancelled_when_engineer_disrupted](#)
[omdurman-rules::src::effects::demolition_cancelled_when_engineer_moved_away](#)
[omdurman-rules::src::effects::demolition_targets_finds_adjacent_fort_and_wall](#)

§6.54 – Forts

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:798 GH:omdurman-rules/src/lib.rs:798	ZocReason	<pre> 796 /// Used by the engine when answering "is this hex in 797 an enemy ZOC?". 798 #[derive(Serialize, Deserialize, Clone, Copy, 799 PartialEq, Eq, Debug)] 800 pub enum ZocReason { 801 /// Normal ZOC: any non-disrupted unit other than 802 an Anglo-Egyptian 803 /// leader (§5.41) projects ZOC into each of its 804 six adjacent hexes. </pre>
omdurman-rules/src/lib.rs:805 GH:omdurman-rules/src/lib.rs:805	Fort	<pre> 803 GunboatVsGunboat, 804 /// Forts project ZOC out of, but not into, an 805 empty fort (§5.44, §6.54). 806 Fort, 807 /// Walled-city ZOC: extends out through walls and 808 gates but not in, 809 /// across a breach in both directions (§5.44). </pre>
omdurman-rules/src/lib.rs:753 GH:omdurman-rules/src/lib.rs:753	UnitState	<pre> 751 /// rather than one big enum. 752 #[derive(Serialize, Deserialize, Clone, Copy, 753 PartialEq, Eq, Debug, Default)] 754 pub struct UnitState { 755 /// Reference table: "Disrupted units: no ZOC; may 756 not move; may not fire 757 /// offensively or defensively; may not melee; are 758 turned face up at the </pre>

omdurman-rules/src/lib.rs:896 GH:omdurman-rules/src/lib.rs:896	FireAttack	<pre> 894 /// modifiers (rulebook \$6). 895 #[derive(Serialize, Deserialize, Clone, Debug, PartialEq, Eq)] 896 pub struct FireAttack { 897 pub firing_player: Player, 898 pub phase: Phase, </pre>
omdurman-rules/src/lib.rs:911 GH:omdurman-rules/src/lib.rs:911	FireAttack::net_modifier	<pre> 909 impl FireAttack { 910 /// Sum of all fire modifiers applied to this attack (rulebook \$6.24). 911 pub fn net_modifier(&self) -> i16 { 912 self.modifiers.iter().map(m m.die_modifier()).sum() 913 } </pre>

Covered by tests: [omdurman-rules::src::effects::retreat_before_melee_may_not_land_on_enemy_fort](#)

§6.61 – Only artillery may fire at gunboats; 3+ to sink

implemented

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Only artillery may fire at gunboats. A result of 3 or more on the combat results table is required to sink a gunboat. Any other result is a miss.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:460 GH:omdurman-rules/src/lib.rs:460	WeaponClass	<pre> 458 /// enum so a "spear" unit cannot accidentally fire on the "Howitzer" line. 459 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 460 pub enum WeaponClass { 461 /// Dervish spears and swords -- no ranged fire at all. 462 Melee, </pre>

Covered by tests: [omdurman-rules::src::effects::rifles_may_not_sink_a_gunboat](#)

§6.62 – Only artillery may fire at forts; 2+ to destroy

implemented

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Only artillery may fire at forts. A result of 2 or more on the combat results table is required to eliminate a fort. Any other result is a miss. If the fort contains any enemy units at the instant it is destroyed, one unit is eliminated with the fort.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:460 GH:omdurman-rules/src/lib.rs:460	WeaponClass	<pre> 458 /// enum so a "spear" unit cannot accidentally fire on the "Howitzer" line. 459 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 460 pub enum WeaponClass { 461 /// Dervish spears and swords -- no ranged fire at all. 462 Melee, </pre>

Covered by tests: [omdurman-rules::src::effects::artillery_destroys_fort_on_two_or_better_only](#)

§6.63 – Only artillery may breach wall hexsides; 2+ required

implemented

manual page unknown

Only artillery may fire to breach a wall hexside of Khartoum or the walled city of Omdurman. A result of 2 or more on the combat results table is required to breach a wall. Any other result is a miss. The effect of the breach is to negate the wall hexside for line of sight purposes. Place a “BREACH” marker in an adjacent hex so that the arrow points to the breached hexside. If any enemy units are adjacent to the wall hexside at the instant it is breached, one enemy unit is eliminated.

****6.64) Howitzer fire:**** Five units in the game have howitzer fire capability. These are the five named British gunboats. They may fire their artillery factor as direct fire during the Direct Fire Subphase (see 4 and 6.41) and may then fire the same artillery factor as howitzer fire during the Maxim Second Fire and Howitzer Subphase (see 4 and 6.42). Exception: no howitzer fire is allowed during night game turns. To fire howitzer fire, select any target hex between 4 and 10 hexes from the firing gunboat (ignoring the Line of Sight Table) and roll the ten-sided die twice. The first die roll is the Combat Results Table die roll and the second roll is the impact hex die roll. Refer to the Howitzer Fire Scattergram on the mapsheet for the impact hex. The designated target hex is hit on a roll of 7-10. (see manual for full text)

See also: §6.41, §6.42, §6.64

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:176 GH:omdurman-types/src/lib.rs:176	Breach	<pre>174 /// Breach in a wall (artillery/§6.63 or Royal 175 Engineers/§6.53). ZOC both 176 /// ways; LOS no longer blocked across the hexside. 177 Breach, 178 /// Khor -- gully/wadi. ZOCs do not extend across 179 (§5.44); advance after 180 /// combat may not cross (§6.82).</pre>
omdurman-rules/src/effects.rs:311 GH:omdurman-rules/src/effects.rs:311	ArtilleryBreachWall	<pre>309 /// pre-rolled d10 used for the CRT lookup; range/ 310 LOS are re-derived by the 311 /// engine from the firers and `target`. 312 ArtilleryBreachWall { 313 firers: Vec<UnitId>, 314 target: HexsideRef,</pre>
omdurman-rules/src/effects.rs:4741 GH:omdurman-rules/src/effects.rs:4741	apply_artillery_breach_wall	<pre>4739 /// artillery's CRT roll -- the rulebook specifies the 4740 same "2+ required" 4741 /// threshold for both trigger styles. 4742 pub fn apply_artillery_breach_wall(4743 state: &mut GameState, 4744 firers: &[UnitId],</pre>
omdurman-rules/src/effects.rs:2033 GH:omdurman-rules/src/effects.rs:2033	can_fire_at_wall	<pre>2031 /// range band and resolving the CRT -- this method 2032 only validates one 2033 /// firer at a time. 2034 pub fn can_fire_at_wall(2035 &self, 2036 firer: UnitId,</pre>

Covered by tests: [omdurman-rules::src::unit_profiles::breach_marker_cell_returns_none](#)

§6.64 – Howitzer fire

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/howitzer_scatter.rs:8 GH:omdurman-rules/src/howitzer_scatter.rs:8	ScatterHexDirection	<pre>6 /// the diagram away from the firing player. 7 #[derive(serde::Serialize, serde::Deserialize, Clone, 8 Copy, PartialEq, Eq, Hash, Debug)] 9 pub enum ScatterHexDirection { 10 UpperLeft, 11 UpperRight,</pre>

omdurman-rules/src/howitzer_scatter.rs:36 GH:omdurman-rules/src/howitzer_scatter.rs:36	howitzer_scatter	<pre> 34 /// onto a hex-grid offset oriented away from the firer (see 35 /// `GameState::howitzer_impact_hex`). 36 pub fn howitzer_scatter(impact_roll: DieRoll) -> ScatterHexDirection { 37 let table = crate::tables_data::scattergram_table(); 38 table </pre>
omdurman-rules/src/lib.rs:414 GH:omdurman-rules/src/lib.rs:414	GunboatId	<pre> 412 /// fire; "old" gunboats do not (rulebook §2.32). 413 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 414 pub enum GunboatId { 415 /// One of the five new-type named gunboats with howitzer capability. 416 Named(NamedGunboat), </pre>
omdurman-rules/src/effects.rs:121 GH:omdurman-rules/src/effects.rs:121	HowitzerFire	<pre> 119 /// - CRT result applied to units at impact hex (not the original target). 120 /// - Firers marked as fired. 121 HowitzerFire { 122 attack: FireAttack, 123 combat_results_table_roll: DieRoll, </pre>
omdurman-rules/src/effects.rs:3384 GH:omdurman-rules/src/effects.rs:3384	apply_howitzer_fire	<pre> 3382 3383 /// Validate and apply a howitzer fire attack (scatter path) (rulebook §6.64). 3384 pub fn apply_howitzer_fire(3385 state: &mut GameState, 3386 attack: &FireAttack, </pre>
omdurman-rules/src/effects.rs:1848 GH:omdurman-rules/src/effects.rs:1848	can_fire_at	<pre> 1846 /// modifier in the [`FireAttack`] and is responsible for the LOS gate. 1847 /// (Howitzer fire ignores LOS entirely -- §6.64.) 1848 pub fn can_fire_at(1849 &self, 1850 firer: UnitId, </pre>

Covered by tests:

- omdurman-rules::src::howitzer_scatter::howitzer_on_target_7_to_10
- omdurman-rules::src::howitzer_scatter::howitzer_scatters_below_7
- omdurman-rules::src::howitzer_scatter::howitzer_each_miss_gets_its_printed_hex
- omdurman-rules::src::unit_profiles::named_and_old_gunboats_resolve
- omdurman-rules::src::effects::named_gunboat_has_howitzer
- omdurman-rules::src::effects::named_gunboat_may_fire_howitzer_in_second_subphase
- omdurman-rules::src::effects::named_gunboat_direct_fire_uses_artillery_weapon
- omdurman-rules::src::effects::named_gunboat_no_howitzer_at_night
- omdurman-rules::src::effects::dervish_gunboat_lacks_howitzer

§6.81 – Moving player may fire with all capable units

implicit

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During Offensive Fire phase, the moving player may fire with all of his units capable of firing, up to their maximum range, within the limitations imposed by the rules of combat.

§6.82 – Advance after combat (offensive fire)

implemented

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If an enemy-occupied hex is vacated as a result of offensive fire, friendly units may advance after combat into the vacated hex. To be eligible to advance, the friendly units must have participated in the attack and must have been adjacent to the vacated hex. Note that artillery may not advance, nor may units advance across a wall hexside (except at a gate or breach). Units may never advance after combat across a khor.

File	Symbol	Code Snippet
------	--------	--------------

omdurman-rules/src/effects.rs:195 GH:omdurman-rules/src/effects.rs:195	AdvanceAfterCombat	<pre>193 /// **Postconditions:** Unit position moved to `to`, `vacated_by_combat` 194 /// entry consumed. 195 AdvanceAfterCombat { unit_id: UnitId, to: HexCoord }, 196 197 // -- Unit state changes -----</pre>
omdurman-rules/src/effects.rs:4489 GH:omdurman-rules/src/effects.rs:4489	apply_advance_after_combat	<pre>4487 4488 /// Apply an advance-after-combat for a unit (rulebook §6.82, §7.6). 4489 pub fn apply_advance_after_combat(4490 state: &mut GameState, 4491 unit_id: UnitId,</pre>
omdurman-types/src/lib.rs:219 GH:omdurman-types/src/lib.rs:219	blocks_advance_after_combat	<pre>217 218 /// Whether advance-after-combat may *not* cross this side (§6.82, §7.6). 219 pub fn blocks_advance_after_combat(self) -> bool { 220 matches!(221 self,</pre>
omdurman-rules/src/effects.rs:4084 GH:omdurman-rules/src/effects.rs:4084	can_advance_after_combat	<pre>4082 /// player's unit, not artillery, adjacent to `to`, and `to` now empty. 4083 /// Wall/khor hexside restrictions are not enforced (no hexside map data). 4084 pub fn can_advance_after_combat(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 4085 let unit = self.unit_or_err(unit_id)?; 4086 // §6.7: there is no advance after combat as a result of defensive fire.</pre>

Covered by tests: [omdurman-rules::src::effects::can_advance_after_combat_rejects_wall_hexside](#)
[omdurman-rules::src::effects::can_advance_after_combat_rejects_khor_hexside](#)
[omdurman-rules::src::effects::advance_requires_combat_vacated_hex](#) [omdurman-rules::src::effects::advance_requires_participation](#)

8/8 implemented

§7 – Melee Phase

§7 – Melee Phase (chapter)

implemented

manual page unknown

Melee Phase

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:951 GH:omdurman-rules/src/effects.rs:951	PendingMelee	<pre>949 /// resolution after the reaction window is deterministic and host-ordered (rulebook §7.5). 950 #[derive(Serialize, Deserialize, Clone, Debug)] 951 pub struct PendingMelee { pub attack: MeleeAttack, 952 pub attacker_roll: DieRoll,</pre>

Covered by tests: [omdurman-rules::src::effects::declared_melee_blocks_phase_advance](#)

§7.1 – Melee strength printed on counter

implemented

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The melee strength of all units is printed on the counter. Note that gunboats have no melee strength. Gunboats may neither melee attack nor be melee attacked.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:104 GH:omdurman-rules/src/lib.rs:104	MeleeFactor	<pre>102 /// Every possible value from the annotated counter set is a named variant. 103 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 104 pub enum MeleeFactor { One = 1, 105 Three = 3,</pre>
omdurman-rules/src/lib.rs:115 GH:omdurman-rules/src/lib.rs:115	MeleeFactor::sum	<pre>113 impl MeleeFactor { 114 /// Sum multiple melee factors (rulebook §7.1). 115 pub fn sum<'a>(factors: impl IntoIterator<Item = &'a MeleeFactor>) -> u16 { 116 factors.into_iter().map(f f.value()).sum() 117 }</pre>
omdurman-types/src/lib.rs:916 GH:omdurman-types/src/lib.rs:916	may_be_melee_attacked	<pre>914 915 /// Gunboats neither attack nor are attacked in melee (§7.1). 916 pub fn may_be_melee_attacked(self) -> bool { 917 !matches!(self, UnitKind::Gunboat { .. }) 918 }</pre>

Covered by tests: [omdurman-rules::src::effects::reinforcement_rejected_onto_enemy_occupied_hex](#)

§7.2 – Melee adjacent only, not across wall hexsides

implemented

manual page unknown

Melee simulates the hand-to-hand fighting of the period. Units may melee attack adjacent enemy units only. Units may not melee attack across a wall hexside, but may melee attack through a gate or breach hexside.

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:214 GH:omdurman-types/src/lib.rs:214	blocks_melee	<pre> 212 /// Whether melee may *not* be made across this side (§7.2). Gates and 213 /// breaches are passable to melee. 214 pub fn blocks_melee(self) -> bool { 215 matches!(self, HexsideKind::Wall HexsideKind::ZaribaThornHedge) 216 } </pre>
omdurman-rules/src/effects.rs:2123 GH:omdurman-rules/src/effects.rs:2123	can_melee	<pre> 2121 /// that may be melee-attacked (gunboats may not -- §7.1), and no wall or 2122 /// thorn-hedge hexside blocks the attack (§7.2). 2123 pub fn can_melee(&self, attacker: UnitId, defender_hex: HexCoord) -> Result<(), RuleError> { 2124 let unit = self.unit_or_err(attacker)?; 2125 </pre>

Covered by tests: [omdurman-rules::src::effects::can_melee_gates_phase_adjacency_and_kind](#)
[omdurman-rules::src::effects::can_melee_rejects_wall_hexside](#) [omdurman-rules::src::effects::can_melee_rejects_thorn_hedge_hexside](#)
[omdurman-rules::src::effects::can_melee_allows_gate_hexside](#)

§7.3 – Simultaneous melee combat

implemented

manual page unknown

Melee combat is considered simultaneous, so that units eliminated by melee attacks still get a melee combat die roll.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:144 GH:omdurman-rules/src/effects.rs:144	MeleeCombat	<pre> 142 /// - Winner may advance into vacated hex (§7.6). 143 /// - Victory points awarded for eliminations. 144 MeleeCombat { 145 attack: MeleeAttack, 146 attacker_roll: DieRoll, </pre>
omdurman-rules/src/effects.rs:3741 GH:omdurman-rules/src/effects.rs:3741	apply_melee_combat	<pre> 3739 3740 /// Apply a simultaneous melee combat between two adjacent hexes (rulebook §7). 3741 pub fn apply_melee_combat(3742 state: &mut GameState, 3743 attack: &MeleeAttack, </pre>

Covered by tests: [omdurman-rules::src::effects::melee_resolves_simultaneously](#)

§7.4 – Who may melee attack / defend

implemented

manual page unknown

Only infantry, cavalry, camel units, and Dervish leaders may melee attack. All units (except gunboats — see 7.1) may melee defend.

See also: §7.1

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:905 GH:omdurman-types/src/lib.rs:905	may_melee_attack	<pre> 903 /// Rulebook §7.4 -- only infantry, cavalry, camel and Dervish leaders may 904 /// melee *attack*. All others (except gunboats) may melee *defend* (§7.1). 905 pub fn may_melee_attack(self) -> bool { </pre>

		<pre> 906 matches!(907 self, </pre>
omdurman-types/src/lib.rs:842 GH:omdurman-types/src/lib.rs:842	UnitKind	<pre> 840 /// `Some(UnitKind::Marker)` or `None`. 841 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 842 pub enum UnitKind { 843 /// Foot infantry (§2.3): fire / melee / movement. 844 Infantry { </pre>
omdurman-types/src/lib.rs:691 GH:omdurman-types/src/lib.rs:691	DervishTribe	<pre> 689 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display, strum::EnumIter, 690)] 691 pub enum DervishTribe { 692 Baggara, 693 Jaalin, </pre>
omdurman-rules/src/effects.rs:2123 GH:omdurman-rules/src/effects.rs:2123	can_melee	<pre> 2121 /// that may be melee-attacked (gunboats may not -- §7.1), and no wall or 2122 /// thorn-hedge hexside blocks the attack (§7.2). 2123 pub fn can_melee(&self, attacker: UnitId, defender_hex: HexCoord) -> Result<(), RuleError> { 2124 let unit = self.unit_or_err(attacker)?; 2125 </pre>

Covered by tests: [omdurman-rules::src::effects::can_melee_gates_phase_adjacency_and_kind](#)

§7.5 – Cavalry/camel retreat before melee

implemented

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Cavalry and camel units may retreat two hexes from an infantry melee attack. Note, however, that only one retreat per unit per turn is permitted. Thus, if their retreat places them adjacent to enemy units whose melee attacks have not yet been resolved, those enemy units may elect to attack the retreating unit(s).

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:181 GH:omdurman-rules/src/effects.rs:181	RetreatBeforeMelee	<pre> 179 /// 180 /// **Postconditions:** Unit position moved to `to`. 181 RetreatBeforeMelee { unit_id: UnitId, to: HexCoord }, 182 183 /// An attacking unit advances into a hex vacated by combat (rulebook §6.82 </pre>
omdurman-rules/src/effects.rs:4454 GH:omdurman-rules/src/effects.rs:4454	apply_retreat_before_melee	<pre> 4452 4453 /// Apply a retreat-before-melee for a cavalry/camel unit (rulebook §7.5). 4454 pub fn apply_retreat_before_melee(4455 state: &mut GameState, 4456 unit_id: UnitId, </pre>
omdurman-types/src/lib.rs:922 GH:omdurman-types/src/lib.rs:922	may_retreat_before_melee	<pre> 920 /// Cavalry and camel units may retreat two hexes from an infantry melee 921 /// attack (§7.5). 922 pub fn may_retreat_before_melee(self) -> bool { 923 matches!(self, UnitKind::Cavalry { .. } UnitKind::Camel { .. }) 924 } </pre>
omdurman-rules/src/lib.rs:181 GH:omdurman-rules/src/lib.rs:181	HexDistance	<pre> 179 /// (rulebook §6.22, §7.5). 180 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 181 pub struct HexDistance(u16); 182 183 impl HexDistance { </pre>

omdurman-rules/src/effects.rs:4015 GH:omdurman-rules/src/effects.rs:4015	can_retreat_before_melee	<pre> 4013 /// two hexes away and empty. (Does not verify the 4014 /// the caller offers the retreat only in response 4015 /// to one.) 4015 pub fn can_retreat_before_melee(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 4016 let unit = self.unit_or_err(unit_id)?; 4017 if !matches!(self.phase, Phase::Melee) { </pre>
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Covered by tests: [omdurman-rules::src::effects::retreat_before_melee_only_cavalry_two_hexes](#)
[omdurman-rules::src::effects::retreat_opens_window_only_when_hex_emptyies](#)

§7.6 – Advance after melee

implemented

manual page unknown

If a melee attack eliminates all of the defenders in an adjacent hex, the Dervish player MUST advance into the vacated hex. To be eligible to advance, the Dervish units must have been adjacent to the vacated hex and participated in the melee attack that eliminated the defenders. All surviving eligible Dervish units MUST advance, up to the stacking limit. The Anglo-Egyptian player may advance if desired. Note that only attacking units may advance.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:195 GH:omdurman-rules/src/effects.rs:195	AdvanceAfterCombat	<pre> 193 /// **Postconditions:** Unit position moved to 194 /// `to`; `vacated_by_combat` 195 /// entry consumed. 195 AdvanceAfterCombat { unit_id: UnitId, to: HexCoord }, 196 197 // -- Unit state changes ----- </pre>
omdurman-rules/src/effects.rs:4489 GH:omdurman-rules/src/effects.rs:4489	apply_advance_after_combat	<pre> 4487 4488 /// Apply an advance-after-combat for a unit (rulebook 4489 /// §6.82, §7.6). 4489 pub fn apply_advance_after_combat(4490 state: &mut GameState, 4491 unit_id: UnitId, </pre>
omdurman-rules/src/effects.rs:4084 GH:omdurman-rules/src/effects.rs:4084	can_advance_after_combat	<pre> 4082 /// player's unit, not artillery, adjacent to 4083 /// `to`, and `to` now empty. 4083 /// Wall/khor hexside restrictions are not 4084 /// enforced (no hexside map data). 4084 pub fn can_advance_after_combat(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 4085 let unit = self.unit_or_err(unit_id)?; 4086 // §6.7: there is no advance after combat as a 4086 // result of defensive fire. </pre>

Covered by tests: [omdurman-rules::src::effects::dervish_advance_after_melee_is_mandatory](#)

§7.7 – Melee modifiers

implemented

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To resolve melee, both the attacker and the defender roll on the Combat Results Table and apply the applicable melee modifier to their die roll. The Dervish player receives a +2 melee modifier, the Anglo-Egyptian player receives a +1 melee modifier. No terrain modifiers are applied to melee combat (Exception: Zariba hexsides in the historical scenario and the campaign game, if constructed — see 9.23). Melee losses must be taken from meleeing units first!

See also: §9.23

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:934 GH:omdurman-rules/src/lib.rs:934	MeleeModifier	<pre> 932 933 #[derive(Serialize, Deserialize, Clone, Copy, 934 pub enum MeleeModifier { 935 /// +2 to all Dervish melee rolls (§7.7). 936 DervishStandard,</pre>
omdurman-rules/src/lib.rs:957 GH:omdurman-rules/src/lib.rs:957	MeleeAttack	<pre> 955 /// A melee attack: simultaneous, both sides roll on 956 the Combat Results Table (§7.3, §7.7). 957 #[derive(Serialize, Deserialize, Clone, Debug, 958 pub struct MeleeAttack { 959 pub attacker_player: Player,</pre>
omdurman-rules/src/lib.rs:938 GH:omdurman-rules/src/lib.rs:938	MeleeModifier::Anglo-EgyptianStandard	<pre> 936 DervishStandard,</pre> <pre> 937 /// +1 to all Anglo-Egyptian melee rolls (§7.7). 938 AngloEgyptianStandard,</pre> <pre> 939 /// Inverted to -2 when Dervish units melee-attack 940 across a trench into 941 /// an entrenched defender (§9.232).</pre>
omdurman-rules/src/lib.rs:941 GH:omdurman-rules/src/lib.rs:941	MeleeModifier::DervishVsDefender	<pre> 939 /// Inverted to -2 when Dervish units melee-attack 940 across a trench into 941 /// an entrenched defender (§9.232). 942 DervishVsTrenchedDefender,</pre> <pre> 943 }</pre>
omdurman-rules/src/lib.rs:936 GH:omdurman-rules/src/lib.rs:936	MeleeModifier::DervishStandard	<pre> 934 pub enum MeleeModifier { 935 /// +2 to all Dervish melee rolls (§7.7). 936 DervishStandard,</pre> <pre> 937 /// +1 to all Anglo-Egyptian melee rolls (§7.7). 938 AngloEgyptianStandard,</pre>

Covered by tests: [omdurman-rules::src::effects::melee_resolves_simultaneously](#)
[omdurman-rules::src::effects::melee_modifiers_are_engine_derived_and_mismatches_rejected](#)

2/3 implemented

§8 – Night Game Turns

§8 – Night Game Turns

descriptive

manual page unknown

Night Game Turns

§8.1 – Night effects

implemented

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The effects of night game turns are: a) all Anglo-Egyptian movement allowances are halved (round down), b) there is no Anglo-Egyptian howitzer fire, and c) all fire ranges are halved for both sides (round down, but range 1 stays range 1). Range effects on fire combat are the same as during day game turns. For example, an Anglo-Egyptian infantry unit firing at night will be doubled at range 1, normal at range 2, and may not fire at range 3 or greater.

****8.2) Dervish Desertion Roll:**** Once each campaign game, during the first night turn of the game, the Dervish player rolls one die to see how many of his units desert. The roll is made during the movement phase and the number of deserting Dervish units is equal to 1½ times the roll of one die. The Dervish player may choose which units desert by merely removing them from the mapsheet. The KHALIFA unit, gunboats, artillery units, and forts are the only Dervish units that may not be chosen. No victory points are awarded to the Anglo-Egyptian player for deserting Dervishes.

(see manual for full text)

See also: §8.2

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:150 GH:omdurman-rules/src/lib.rs:150	MovementAllowance::halve	<pre>148 impl MovementAllowance { 149 /// Night movement allowance = halved (round down) (rulebook §8.1, §5.11). 150 pub fn halve(self) -> Self { 151 let v = self.value() / 2; 152 MovementAllowance::try_from(v).expect("halved value always a named variant")</pre>
omdurman-types/src/lib.rs:756 GH:omdurman-types/src/lib.rs:756	DayNight	<pre>754 /// (rulebook §8.1). 755 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 756 pub enum DayNight { 757 Day, 758 Night,</pre>
omdurman-rules/src/range_effects.rs:68 GH:omdurman-rules/src/range_effects.rs:68	night_max_range	<pre>66 67 /// The halved maximum range at night (§8.1): round down, minimum 1. 68 pub fn night_max_range(weapon: WeaponClass, ae: bool) - > u8 { 69 let day = max_day_range(weapon, ae); 70 if day <= 1 { 1 } else { day / 2 }</pre>
omdurman-rules/src/lib.rs:1429 GH:omdurman-rules/src/lib.rs:1429	effective_movement_at_night	<pre>1427 /// Apply night-turn movement halving for Anglo- Egyptian units (§8.1): all 1428 /// Anglo-Egyptian movement allowances are halved (round down). 1429 pub fn effective_movement_at_night(1430 allowance: MovementAllowance, 1431 player: Player,</pre>

Covered

by tests: `omdurman-rules::src::range_effects::night_max_ranges` `omdurman-rules::src::range_effects::night_max_ranges_remaining`
`omdurman-rules::src::range_effects::ae_rifle_at_night_matches_rulebook_example`
`omdurman-rules::src::range_effects::max_day_range_all_combos`
`omdurman-rules::src::effects::night_movement_overlay_allowance_halved`

§8.2 – Dervish Desertion Roll

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:223 GH:omdurman-rules/src/effects.rs:223	DervishDesertion	<pre>221 /// the effect. The Khalifa, gunboats, artillery, and forts may not be 222 /// chosen. 223 DervishDesertion { 224 roll: DieRoll, 225 deserters: Vec<UnitId>,</pre>
omdurman-rules/src/turn_track.rs:63 GH:omdurman-rules/src/turn_track.rs:63	DervishDesertion	<pre>61 None, 62 /// Dervish desertion roll (§8.2) -- occurs on the first night turn. 63 DervishDesertion, 64 /// Dervish reinforcements are available. 65 DervishReinforcements,</pre>
omdurman-rules/src/turn_track.rs:60 GH:omdurman-rules/src/turn_track.rs:60	TurnEvent	<pre>58 /// Special events that occur on specific turns (rulebook §8.2, §9.112, §9.113). 59 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 60 pub enum TurnEvent { 61 None, 62 /// Dervish desertion roll (§8.2) -- occurs on the first night turn.</pre>

Covered

by

tests:

`omdurman-rules::src::effects::desertion_count_is_floor_one_and_a_half` `omdurman-rules::src::turn_track::desertion_on_first_night`
`omdurman-rules::src::effects::desertion_roll_required_before_first_night_movement_ends`

 22/33 implemented

§9 – The Scenarios

§9 – The Scenarios

descriptive

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The Scenarios

§9.1 – The Campaign Game

descriptive

manual page unknown

The Campaign Game

Covered

by

tests:

omdurman-app::src::scenario_setup::campaign_has_no_fixed_placements omdurman-app::src::tests::scenario_maps_to_board

§9.2 – The Historical Scenario

descriptive

manual page unknown

The Historical Scenario

Players should note that the historical scenario is an exercise in futility for the Dervish player. It is, however, an interesting demonstration of the absolute imbecility of the Khalifa's generalship and vividly shows the superiority of entrenched firepower over traditional tribal arms in the colonial period.

Covered

by

tests:

omdurman-app::src::tests::start_game_scenario_selects_board omdurman-rules::src::effects::remove_deployed_unit_happy_path

§9.3 – Bonus Game: Fall of Khartoum

out-of-scope

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Bonus Game: FALL OF KHARTOUM

Covered by tests: omdurman-rules::src::effects::remove_deployed_unit_happy_path

§9.11 – Set Up (Campaign)

out-of-scope

manual page unknown

Set Up

§9.12 – Scenario Length (Campaign)

implemented

manual page unknown

Scenario Length

6:00 am, Sept. 1 through 8:00 am, Sept. 3, 22 Game Turns.

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:11 GH:omdurman-rules/src/turn_track.rs:11	GameTime	<pre> 9 /// starts at one of these twelve times. 10 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 11 pub enum GameTime { 12 SixAM, 13 EightAM,</pre>
omdurman-rules/src/turn_track.rs:47 GH:omdurman-rules/src/turn_track.rs:47	TurnEntry	<pre> 45 /// A single entry on the Turn Record Track (rulebook \$9.12, \$9.22). 46 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 47 pub struct TurnEntry { 48 /// 1-based turn number. 49 pub turn: u8,</pre>
omdurman-rules/src/turn_track.rs:95 GH:omdurman-rules/src/turn_track.rs:95	CAMPAIGN_TURN_TRACK	<pre> 93 /// is turn 9, which carries the once-per-game Dervish Desertion Roll (§8.2) -- 94 /// the printed track prints "Dervish Desertion Roll / NIGHT" on that cell. 95 pub const CAMPAIGN_TURN_TRACK: [TurnEntry; 22] = [96 // Row 1, left->right: Sept 1, 6 am -> 8 pm, then the first NIGHT. 97 entry(1, SixAM, Day, TurnEvent::None),</pre>
omdurman-rules/src/turn_track.rs:242 GH:omdurman-rules/src/turn_track.rs:242	TurnLabel	<pre> 240 /// printed cell; `Blank` is for unused positions in the 9x3 grid. 241 #[derive(Serialize, Deserialize, Debug, Clone, Copy, PartialEq, Eq)] 242 pub enum TurnLabel { 243 Blank, 244 /// "SEPT. 1\n6:00 am" -- day header plus time (turn 1).</pre>

Covered

by tests: `omdurman-rules::src::turn_track::campaign_track_22_turns` `omdurman-rules::src::turn_track::desertion_on_first_night` `omdurman-rules::src::turn_track::campaign_track_label_and_day_night_agree` `omdurman-rules::src::turn_track::game_time_display_all_variants` `omdurman-rules::src::turn_track::turn_label_display` `omdurman-rules::src::turn_track::turn_label_out_of_range_is_none` `omdurman-rules::src::effects::game_over_after_campaign_turns`

§9.13 – Special Rules (Campaign)

descriptive

manual page unknown

Special Rules

None.

§9.14 – Victory Conditions (Campaign)

implemented

manual page unknown

Victory Conditions

The Mahdi's Tomb in Omdurman was not only the tallest structure in the entire Sudan in 1898, it was also a Dervish holy shrine. Its loss or destruction would be a severe blow to the Mahdist cause. It is accordingly assigned 25 victory points which are awarded to the player who controls it at the conclusion of play. The Dervish player controls it at the start of play. As a tactical note, the Anglo-Egyptian player will find a decisive victory almost impossible unless he takes the Mahdi's Tomb from the Dervish player. To take the Tomb hex, it must be occupied by one British leader plus any one non-“Friendlies” Anglo-Egyptian combat unit (both undisrupted) at the conclusion of play.

Additional victory points are awarded as follows:

****Dervish Player receives:****

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1083 GH:omdurman-rules/src/lib.rs:1083	VpSource	<pre> 1081 /// the manual and the engine. 1082 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1083 pub enum VpSource { 1084 // ----- Anglo-Egyptian player receives: 1085 // Mahdi's Tomb control at conclusion of play (§9.14). </pre>
omdurman-rules/src/lib.rs:1109 GH:omdurman-rules/src/lib.rs:1109	VpSource::points	<pre> 1107 impl VpSource { 1108 /// VP awarded to `who_scores()` (rulebook §9.14). 1109 pub fn points(self) -> VictoryPoints { 1110 match self { 1111 VpSource::MahdisTomb => VictoryPoints::new(25), </pre>
omdurman-rules/src/lib.rs:1124 GH:omdurman-rules/src/lib.rs:1124	VpSource::who_scores	<pre> 1122 1123 /// Which player receives these victory points (rulebook §9.14). 1124 pub fn who_scores(self) -> Player { 1125 match self { 1126 VpSource::MahdisTomb </pre>
omdurman-rules/src/lib.rs:1159 GH:omdurman-rules/src/lib.rs:1159	VictoryLedger	<pre> 1157 /// Cumulative victory ledger for one scenario (rulebook §9.14). 1158 #[derive(Serialize, Deserialize, Clone, Debug, Default)] 1159 pub struct VictoryLedger { 1160 pub events: Vec<VpEvent>, 1161 } </pre>
omdurman-rules/src/lib.rs:1165 GH:omdurman-rules/src/lib.rs:1165	VpEvent	<pre> 1163 /// A single victory-point scoring event (rulebook §9.14). 1164 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1165 pub struct VpEvent { 1166 pub turn: GameTurnIndex, 1167 pub source: VpSource, </pre>
omdurman-rules/src/lib.rs:1172 GH:omdurman-rules/src/lib.rs:1172	VictoryLedger::total_for	<pre> 1170 impl VictoryLedger { 1171 /// Total victory points earned by a given player (rulebook §9.14). 1172 pub fn total_for(&self, player: Player) -> VictoryPoints { 1173 VictoryPoints(1174 self.events </pre>
omdurman-rules/src/lib.rs:1184 GH:omdurman-rules/src/lib.rs:1184	VictoryLedger::superiority	<pre> 1182 /// Net superiority: positive = Anglo-Egyptian ahead, negative = Dervish ahead 1183 /// (rulebook §9.14). 1184 pub fn superiority(&self) -> VictoryPoints { 1185 VictoryPoints(1186 self.total_for(Player::AngloEgyptian).value() - self.total_for(Player::Dervish).value(), </pre>
omdurman-rules/src/lib.rs:1204 GH:omdurman-rules/src/lib.rs:1204	CampaignVictoryLevel	<pre> 1202 /// Campaign-game victory levels (§9.14). 1203 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1204 pub enum CampaignVictoryLevel { 1205 Draw, 1206 Marginal(Player), </pre>
omdurman-rules/src/lib.rs:1213 GH:omdurman-	CampaignVictoryLevel::f	<pre> 1211 impl CampaignVictoryLevel { 1212 /// Assign a level from the net superiority (§9.14). 1213 pub fn from_superiority(s: VictoryPoints) -> Self </pre>

rules/src/ lib.rs:1213		<pre>{ 1214 let net = s.0; 1215 // Positive -> Anglo-Egyptian thresholds: 15/30/50</pre>
omdurman- rules/src/ effects.rs:5503 GH:omdurman- rules/src/ effects.rs:5503	score_elimination	<pre>5501 /// elimination under `cause`. The owner is derived 5502 /// so unlike the historical signature there is no 5503 /// caller-supplied player. 5504 pub fn score_elimination(state: &mut GameState, 5505 unit_id: UnitId, cause: ElimCause) { 5506 if let Some(unit) = state.find_unit(unit_id) { 5507 let identity = unit.profile.identity;</pre>
omdurman- rules/src/ lib.rs:225 GH:omdurman- rules/src/lib.rs:225	VictoryPoints	<pre>223 /// (rulebook §9.14). 224 #[derive(Serialize, Deserialize, Clone, Copy, 225 PartialEq, Eq, PartialOrd, Ord, Debug, Default)] 226 pub struct VictoryPoints(i32); 227 impl VictoryPoints {</pre>

Covered by tests: omdurman-rules::src::effects::friendlyes_bank_scores_by_side
omdurman-rules::src::effects::mahdis_tomb_not_scored_without_a_leader
omdurman-rules::src::effects::mahdis_tomb_scores_for_angular_egyptian_when_held omdurman-rules::src::lib::vp_source_attributes

§9.21 – Set Up (Historical)

out-of-scope
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Set Up

§9.22 – Scenario Length (Historical)

implemented
manual page unknown

Scenario Length

6:00 am, September 2 through 12:00 noon, September 2. Four game turns.

File	Symbol	Code Snippet
omdurman- rules/src/turn_track.rs:132 GH:omdurman- rules/src/ turn_track.rs:132	HISTORICAL_TURN	<pre>130 131 /// Historical scenario track (§9.22 -- 4 turns, Sept 2 132 6:00 am -> 12:00 pm). 133 pub const HISTORICAL_TURN_TRACK: [TurnEntry; 4] = [134 TurnEntry { 135 turn: 1,</pre>

Covered by tests: omdurman-rules::src::turn_track::historical_turn_all_four_turns

§9.23 – Special Rule: The Zariba

descriptive
manual page unknown

Special Rule: “The Zariba”

9.231) Thorn hedge hexides: –2 to die roll on all Dervish fire attacks; may not melee across in either direction; may not advance after combat across in either direction.

****9.232) Trench hexsides:**** -4 to die roll on all Dervish fire attacks vs. entrenched units only; -2 (instead of +2) melee modifier to Dervish units melee attacking an entrenched unit; entrenched units may be fired “over” in both directions (i.e. they do not block line of sight); units are considered “entrenched” if they are directly adjacent to (and on the Nile River side of) a trench hexside.

(see manual for full text)

See also: §9.231, §9.232

§9.24 – Victory Conditions (Historical)

implemented

manual page unknown

Victory Conditions

Victory Levels are based solely on eliminating enemy units while conserving your own force as much as possible.

| Victory Level | Anglo-Egyptian Player (Dervish units eliminated) | Dervish Player (Anglo-Egyptian units eliminated) |
| |—|—|—| | 5 — DECISIVE | 100+ | 30+ | | 4 — STRATEGIC | 60–99 | 15–29 | | 3 — TACTICAL | 45–59 | 10–14 | | 2 — MARGINAL | 30–44 | 5–9 | | 1 — DRAW | 0–29 | 0–4 |

The lower value victory level is then subtracted from the higher level to determine a player’s net victory. For example, if the Anglo-Egyptian player eliminates 104 Dervish units (decisive victory) but loses 18 units doing it (Dervish Strategic), the Anglo-Egyptian player only nets out with a draw (decisive worth 5 minus strategic worth 4 = 1, draw).

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1243 GH:omdurman-rules/src/lib.rs:1243	HistoricalVictoryLevel	<pre>1241 /// draw"). 1242 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Debug)] 1243 pub enum HistoricalVictoryLevel { 1244 Draw = 1, 1245 Marginal = 2,</pre>

Covered by tests:
[omdurman-rules::src::lib::historical_victory_level_for_dervish](#) [omdurman-rules::src::lib::historical_victory_level_for_anglo_egyptian](#)

§9.31 – Bonus game map

out-of-scope

manual page unknown

Only the small FALL OF KHARTOUM scenario map is used for this game.

§9.32 – Set Up (Bonus)

out-of-scope

manual page unknown

Set Up

§9.33 – Scenario Length (Bonus)

implemented

manual page unknown

Scenario Length

Variable, see victory conditions (9.35). Rarely lasts five turns.

See also: §9.35

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:172 GH:omdurman-rules/src/turn_track.rs:172	FALL_OF_KHARTOUM	<pre>170 /// (the rulebook fixes none); only `day_night` is rule-bearing (night halves 171 /// Anglo-Egyptian movement and ranges and bars howitzer fire, §8.1). 172 pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] = [173 TurnEntry { 174 turn: 1,</pre>

Covered by tests: [omdurman-rules::src::turn_track::fall_of_khartoum_turn_one_is_night](#)
[omdurman-rules::src::turn_track::fall_of_khartoum_turns_3_to_8_are_day](#)

§9.34 – Special Rules (Bonus)

out-of-scope

manual page unknown

Special Rules

§9.35 – Victory Conditions (Bonus)

implemented

manual page unknown

Victory Conditions

Victory is determined by how many turns it takes the Dervish player to eliminate the GORDON leader unit and how many Dervish units are eliminated:

- Dervish decisive: eliminate GORDON turn four or sooner.
- Dervish tactical: eliminate GORDON turn five.
- Dervish marginal: eliminate GORDON turn six.
- British marginal: GORDON survives end of turn six.
- British tactical: GORDON survives end of turn seven.
- British decisive: GORDON survives end of turn eight.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1361 GH:omdurman-rules/src/lib.rs:1361	FoKVictoryLevel::resolve	<pre>1359 /// rulebook: GORDON dies turn 5 (tactical) with 24 Dervish losses (-2 1360 /// levels) nets a British marginal. 1361 pub fn resolve(gordon_died_turn: Option<u8>, scenario_end_turn: u8, dervish_lost: i16) -> Self { 1362 let base = Self::base(gordon_died_turn, scenario_end_turn); 1363 let base_idx = Self::LADDER</pre>

Covered by tests: [omdurman-rules::src::lib::fok_victory_level_worked_example](#) [omdurman-rules::src::lib::fok_victory_level_gordon_died_early](#)
[omdurman-rules::src::lib::fok_victory_level_late_gordon_death](#) [omdurman-rules::src::lib::fok_victory_level_gordon_survived](#)

§9.111 – Dervish set up (Campaign)

implemented

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Dervish player sets up first, moves second.

- Isa Zachneih infantry unit: anywhere on the east bank, in or south of El Debeba.
- KHALIFA ABDULLAH: in the walled city of Omdurman, in either palace hex.
- 3 artillery units, and all Taiasha units: anywhere in the walled city of Omdurman.
- 17 forts: anywhere on the mapsheet south of the Khor Shambat on the west bank, and/or south of all Halfaya hut hexes on the east bank and Nile River islands.
- 2 gunboats: any south edge Nile River hexes.

****9.112) Dervish reinforcements:**** all reinforcements enter on the west edge of the mapsheet, south of the Khor Shambat. Each unit pays the terrain cost of the hex through which it enters, no matter how many units enter through

(see manual for full text)

See also: §9.112

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1061 GH:omdurman-rules/src/effects.rs:1061	setup_complete	<pre>1059 /// currently shares the same "both sides 1060 deployed" gate; when a scenario 1061 /// needs a different minimum, branch on 1062 `self.scenario` here. 1063 pub fn setup_complete(&self) -> Result<(), RuleError> { let has = player { self.units</pre>

Covered by tests: [omdurman-rules::src::unit_profiles::hadendowa_first_cell_is_isa_zachneih](#)
[omdurman-rules::src::effects::campaign_deployment_is_boat_land_exclusive](#)
[omdurman-rules::src::effects::campaign_setup_rejects_non_initial_force](#)

§9.112 – Dervish reinforcements (Campaign)

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:215 GH:omdurman-rules/src/effects.rs:215	PlaceReinforcements	<pre>213 // -- Reinforcement / placement 214 ----- 215 /// Place reinforcements onto the map (rulebook 216 §9.112, §9.113). 217 PlaceReinforcements(Vec<UnitPlacement>), 218 // -- Scenario-specific 219 -----</pre>
omdurman-rules/src/effects.rs:4859 GH:omdurman-rules/src/effects.rs:4859	apply_place_reinforceme	<pre>4857 4858 /// Place reinforcements onto the map (rulebook 4859 §9.112, §9.113). 4860 pub fn apply_place_reinforcements(4861 state: &mut GameState, 4862 placements: &[UnitPlacement],</pre>
omdurman-rules/src/reinforcements.rs:69 GH:omdurman-rules/src/reinforcements.rs:69	dervish_campaign_sched	<pre>67 /// All reinforcements enter on the west edge, south of 68 the Khor Shambat. 69 /// Each unit pays terrain cost of the hex it enters 70 through. 71 pub fn dervish_campaign_schedule() -> ReinforcementSchedule { ReinforcementSchedule { player: Player::Dervish,</pre>

omdurman-types/src/lib.rs:502 GH:omdurman-types/src/lib.rs:502	Location	<pre> 500 /// Named map landmarks (rulebook mapsheet, §9.111, 501 #[derive(Serialize, Deserialize, Clone, Copy, 502 pub enum Location { 503 FortMakran, 504 NorthFort, </pre>
omdurman-types/src/lib.rs:582 GH:omdurman-types/src/lib.rs:582	SetupLetter	<pre> 580 /// Each letter marks a specific hex where a Dervish 581 #[derive(Serialize, Deserialize, Clone, Copy, 582 pub enum SetupLetter { 583 Y, 584 K, </pre>
omdurman-types/src/lib.rs:714 GH:omdurman-types/src/lib.rs:714	Faction	<pre> 712 /// `Some(BrigadeId::friendlies())`. 713 #[derive(Serialize, Deserialize, Clone, Copy, 714 pub enum Faction { 715 Dervish { 716 tribe: DervishTribe, </pre>

Covered by tests:

- [omdurman-rules::src::reinforcements::dervish_schedule_has_three_waves](#)
- [omdurman-rules::src::reinforcements::dervish_wave_one_has_baggaara_and_three_leaders](#)
- [omdurman-rules::src::reinforcements::dervish_wave_two_has_hadendowa](#)
- [omdurman-rules::src::reinforcements::dervish_wave_three_is_all_remaining](#)
- [omdurman-rules::src::reinforcements::wave_for_turn_returns_correct_wave](#)
- [omdurman-rules::src::effects::campaign_reinforcements_gate_by_wave](#)
- [omdurman-rules::src::effects::campaign_setup_rejects_non_initial_force](#)

§9.113 – Anglo-Egyptian reinforcements (Campaign)

[implemented](#)

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The Anglo-Egyptian player moves first. There are no Anglo-Egyptian units on the mapsheet at start. The GORDON unit is not used in this scenario.

- The leader units KITCHENER, GATACRE, and HUNTER may enter anytime during the first four game turns and do not count against the 12 units per turn limit. All three leaders must be in play by the end of turn four!
- All gunboats enter through any north edge Nile River hex, paying one movement point for the first hex entered. The “Friendlies” brigade enters through the Abu Alim hut hex on the east bank, paying eight movement points per unit. All other Anglo-Egyptian units enter through the west bank “ANGLO-EGYPTIAN ENTRANCE AREA”, each unit paying one movement point to enter the mapsheet.
- Turn 1) Any three gunboats; “Friendlies” brigade; Egyptian Cavalry; Horse Artillery; and two infantry brigades

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/reinforcements.rs:126 GH:omdurman-rules/src/reinforcements.rs:126	anglo_egyptian_campaig	<pre> 124 /// - "Friendlies" enter via Abu Alim hut on the east 125 /// - All other AE units enter via the Anglo-Egyptian 126 pub fn anglo_egyptian_campaign_schedule() -> 127 ReinforcementSchedule { 128 let free_leaders = vec![129 CampaignLeader::British(BritishLeader::Kitchener), </pre>

Covered by tests:

- [omdurman-rules::src::reinforcements::anglo_egyptian_schedule_has_four_waves](#)
- [omdurman-rules::src::reinforcements::anglo_egyptian_leaders_available_each_wave](#)
- [omdurman-rules::src::reinforcements::anglo_egyptian_turn_four_is_all_remaining](#)
- [omdurman-rules::src::effects::campaign_reinforcement_cap_and_double_entry](#)

omdurman-rules::src::effects::campaign_gunboats_quota_three_per_turn
omdurman-rules::src::effects::reinforcement_rejected_onto_enemy_occupied_hex
omdurman-rules::src::effects::campaign_setup_rejects_non_initial_force

§9.211 – Anglo-Egyptian set up first, moves second (Historical)

implemented

manual page unknown

The Anglo-Egyptian player sets up first, and moves second:

- Not in play: GORDON leader unit, “Friendlies” brigade.
- Gunboats start in any Nile River hexes adjacent to the Zariba.
- Camel Corps, Egyptian Cavalry, and Horse Artillery start in the village of Kerreri hut hexes.
- All remaining Anglo-Egyptian units set up in the 13 hexes of the Zariba.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:2979 GH:omdurman-rules/src/effects.rs:2979	first_player	<pre>2977 2978 /// The player who moves first in a scenario (§4, 2979 pub fn first_player(scenario: Scenario) -> Player { 2980 match scenario { 2981 Scenario::Campaign => Player::AngloEgyptian,</pre>

Covered by tests: omdurman-rules::src::effects::historical_setup_rejects_not_in_play_units

§9.212 – Dervish set up (Historical) – deployment zones and leader-to-hex mapping

implemented

manual page unknown

The Dervish player sets up second, and moves first.

- Not in play: Isa Zachneih, gunboats, and forts.
- All Dervish units must be set up out of the line of sight of all Anglo-Egyptian units.
- Dervish leaders start on the lettered hexes:
 - A: Ali Wad Helu
 - D: Sheik El Din
 - Y: Yakub
 - K: Khalifa Abdullah
 - S: Sherif
 - O: Osman Digna

“All Dervish units must be set up out of the line of sight of all Anglo-Egyptian units.” (see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1171 GH:omdurman-rules/src/effects.rs:1171	in_deployment_zone	<pre>1169 /// plan / UI rather than this hex predicate. 1170 Documented, not silently 1171 /// dropped. 1172 pub fn in_deployment_zone(&self, player: Player, 1173 hex: HexCoord, is_boat: bool) -> bool { 1174 // No board attached -> permissive (unit 1175 tests, unbound session). 1176 if self.board.terrain.is_empty() {</pre>
omdurman-rules/src/lib.rs:373 GH:omdurman-rules/src/lib.rs:373	Dervish- Leader::setup_letter	<pre>371 /// (§9.212): A→Ali Wad Helu, D→Sheik El Din, 372 Y→Yakub, K→Khalifa Abdullah, 373 S→Sherif, O→Osman Digna. Inverse of 374 ['dervish_leader_for_setup_letter']. 375 pub fn setup_letter(self) -> SetupLetter { 376 match self {</pre>

		<pre> 375 DervishLeader::AliWadHelu => SetupLetter::A, </pre>
omdurman-rules/src/lib.rs:389 GH:omdurman-rules/src/lib.rs:389	dervish_leader_for_setup	<pre> 387 /// inherent impl here, so the mapping is a free 388 function -- the bijective 389 pub fn dervish_leader_for_setup_letter(letter: SetupLetter) -> DervishLeader { 390 match letter { 391 SetupLetter::A => DervishLeader::AliWadHelu, </pre>

Covered by tests:

- [omdurman-rules::src::effects::deploy_rejected_outside_zone](#)
- [omdurman-rules::src::unit_profiles::embedded_leaders_resolve_from_their_host_section](#)
- [omdurman-app::src::scenario_setup::historical_places_all_six_leaders_when_anchors_present](#)
- [omdurman-app::src::scenario_setup::missing_anchor_is_reported_not_dropped_silently](#)
- [omdurman-rules::src::effects::setup_letter_dervish_leader_roundtrip](#)
- [omdurman-rules::src::effects::setup_letter_to_dervish_leader_known_values](#)
- [omdurman-rules::src::effects::historical_setup_rejects_not_in_play_units](#)

§9.231 – Thorn hedge hexsides

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:859 GH:omdurman-rules/src/lib.rs:859	ZaribaThornHedge	<pre> 857 Terrain(i16), 858 /// -2 thorn-hedge defensive modifier (§9.231). 859 ZaribaThornHedge, 860 /// -4 trench defensive modifier (§9.232). Only applies vs. "entrenched" 861 /// units (those Nile-side of the trench hexside). </pre>
omdurman-types/src/lib.rs:184 GH:omdurman-types/src/lib.rs:184	ZaribaThornHedge	<pre> 182 Crest, 183 /// Historical-scenario thorn-hedge segment of the Zariba (§9.231). 184 ZaribaThornHedge, 185 /// Historical-scenario trench segment of the Zariba (§9.232). 186 ZaribaTrench, </pre>
omdurman-rules/src/board.rs:267 GH:omdurman-rules/src/board.rs:267	has_zariba_thorn_hedge	<pre> 265 /// hexside on its perimeter – i.e. whether the ZaribaThornHedge modifier 266 /// applies (§9.231). 267 pub fn has_zariba_thorn_hedge(&self, hex: HexCoord) -> bool { 268 for n in hex.neighbors() { 269 if let Some(kind) = self.hexside_between(hex, n) </pre>

Covered by tests: [omdurman-rules::src::effects::zariba_fire_penalties_apply_to_dervish_fire_only](#)

§9.232 – Trench hexsides

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:862 GH:omdurman-rules/src/lib.rs:862	ZaribaTrenchEntrenched	<pre> 860 /// -4 trench defensive modifier (§9.232). Only applies vs. "entrenched" 861 /// units (those Nile-side of the trench hexside). 862 ZaribaTrenchEntrenched, 863 } 864 </pre>

omdurman-types/src/lib.rs:186 GH:omdurman-types/src/lib.rs:186	ZaribaTrench	<pre> 184 ZaribaThornHedge, 185 /// Historical-scenario trench segment of the Zariba (§9.232). 186 ZaribaTrench, 187 /// One of the two end hexsides of a Zariba trench segment that connect to 188 /// the Nile River (§9.233). Units may only enter/ leave the Zariba via </pre>
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Covered by tests: [omdurman-rules::src::effects::trench_entrenched_units_take_trench_modifiers](#)

§9.233 – Zariba entry/exit costs

implemented

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Units may only enter and/or leave the Zariba via the two end hexsides that connect to the Nile River, paying +2 movement points to cross (Exception: advance after combat across an entrenched hexside).

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:236 GH:omdurman-types/src/lib.rs:236	blocks_movement	<pre> 234 /// `omdurman-rules`. The trench *end* variants are therefore intentionally 235 /// not blocking. 236 pub fn blocks_movement(self) -> bool { 237 matches!(238 self, </pre>
omdurman-rules/src/board.rs:282 GH:omdurman-rules/src/board.rs:282	zariba_entry_surcharge	<pre> 280 /// movement points to cross"). Returns 2 when the edge between `from` and 281 /// `to` is one of the two trench ends, else 0. 282 pub fn zariba_entry_surcharge(&self, from: HexCoord, to: HexCoord) -> i16 { 283 match self.hexside_between(from, to) { 284 Some(k) if k.is_zariba_trench_end() => 2, </pre>
omdurman-rules/src/effects.rs:1666 GH:omdurman-rules/src/effects.rs:1666	movement_cost_for	<pre> 1664 /// 1665 /// §5.42: entering or leaving an enemy ZOC adds no MP cost. 1666 pub fn movement_cost_for(1667 &self, 1668 unit: &UnitPlacement, </pre>
omdurman-rules/src/effects.rs:3191 GH:omdurman-rules/src/effects.rs:3191	apply_move_unit	<pre> 3189 /// the true terrain cost (§5.11) and enforces gunboat upstream/downstream 3190 /// allowances (§5.24); otherwise it falls back to the caller-supplied `cost`. 3191 pub fn apply_move_unit(3192 state: &mut GameState, 3193 unit_id: UnitId, </pre>

Covered by tests: [omdurman-rules::src::effects::zariba_end_hexside_costs_extra_mp](#)
[omdurman-rules::src::effects::zariba_thorn_hedge_blocks_movement](#)

§9.321 – British set up (Bonus)

implemented

manual page unknown

The British player sets up first, moves second:

- General GORDON leader unit in the palace.
- Two old style (unnamed) gunboats in any Nile River hexes.
- Set up in any building or hut hexes of Khartoum, Forts Makran and/or Buri, and/or adjacent to any wall hex:

- ▶ one Egyptian Battalion artillery unit
- ▶ two British infantry units (represents Caucasian troops)
- ▶ three Egyptian infantry units (represents Cairo “Bazouks”)
- ▶ four Sudan infantry units (represents Sudanese blacks)
- ▶ four “Friendlies” units (represents the Shaggyeh)

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/scenario_setup.rs:42 GH:omdurman-rules/src/scenario_setup.rs:42	FALL_OF_KHARTOUM	<pre> 40 /// North Fort uses a campaign HadendowaForts counter (one of the spare fort 41 /// sprites). 42 pub const FALL_OF_KHARTOUM_SETUP: &[FixedPlacement] = &[43 FixedPlacement { 44 section: SectionName::BritishBoats,</pre>

Covered by tests:

- omdurman-rules::src::effects::confirm_ready_rejected_below_scenario_target
- omdurman-app::src::scenario_setup::fall_of_khartoum_places_gordon_in_the_palace
- omdurman-app::src::scenario_setup::fall_of_khartoum_reports_missing_palace
- omdurman-app::src::scenario_setup::fall_of_khartoum_fort_landmarks_sit_at_the_correct_hexes
- omdurman-rules::src::effects::fok_ae_gunboat_deploys_only_on_nile
- omdurman-rules::src::effects::fok_ae_land_unit_rejected_on_nile
- omdurman-rules::src::effects::fok_setup_complete_requires_full_oob
- omdurman-rules::src::effects::deploy_via_real_sprite_resolution_matches_engine
- omdurman-rules::src::effects::british_boats_named_vs_old_gunboat_detection
- omdurman-rules::src::effects::fok_order_of_battle_british
- omdurman-rules::src::effects::fok_order_of_battle_dervish
- omdurman-rules::src::effects::fok_caps_bind_across_counter_variants

§9.322 – Dervish enters turn one (Bonus)

implemented

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Dervish player moves first: enters turn one through any hexes on the south or east edge of the map.

- 32 Mulazmin units (represents combined forces of Wad El Nejumi, Abu Girgeh, and Sheik El Obeid)
- 2 Hadendowa; 6 Kehena; 5 Degheim (represents Mahdi’s combined west bank forces)
- 3 Dervish artillery units.

File	Symbol	Code Snippet
omdurman-rules/src/unit_profiles.rs:332 GH:omdurman-rules/src/unit_profiles.rs:332	ali_wad_helu	<pre> 330 /// (3-6-9) -- the Degheim force of §9.322, printed on Baggara-backed 331 /// sprites. 332 pub fn ali_wad_helu(col: u32, row: u32) -> Option<Classification> { 333 match (col, row) { 334 (0, 0) => dervish_leader(DervishLeader::AliWadHelu),</pre>
omdurman-types/src/lib.rs:806 GH:omdurman-types/src/lib.rs:806	sections_for_picker	<pre> 804 /// provides the 3 artillery, and HadendowaForts supplies the 805 /// Dervish-controlled North Fort sprite (§9.344). 806 pub fn sections_for_picker(self) -> Option<&'static [SectionName]> { 807 match self { 808 Scenario::Campaign Scenario::Historical => None,</pre>

Covered by tests:

- omdurman-rules::src::unit_profiles::ali_wad_helu_block_resolves_leader_and_degelim_tribes
- omdurman-rules::src::effects::fok_setup_complete_requires_full_oob
- omdurman-rules::src::effects::fok_dervish_land_unit_rejected_on_nile

omdurman-types::src::lib::fok_picker_allowlist_has_dervish_entry_force_blocks

omdurman-rules::src::effects::fok_order_of_battle_dervish omdurman-rules::src::effects::fok_dervish_east_edge_on_diamond_board

§9.341 – Turn 1 is always a night turn (Bonus)

implemented

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Turn 1 is always a night turn (see 8.1).

See also: §8.1

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:172 GH:omdurman-rules/src/turn_track.rs:172	FALL_OF_KHARTOUM	<pre>170 /// (the rulebook fixes none); only `day_night` is rule-bearing (night halves 171 /// Anglo-Egyptian movement and ranges and bars howitzer fire, §8.1). 172 pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] = [173 TurnEntry { 174 turn: 1,</pre>

Covered by tests: omdurman-rules::src::turn_track::fall_of_khartoum_turn_one_is_night

§9.342 – All hexes are playable, including half hexes (Bonus)

implemented

manual page unknown

All hexes are playable, including hexes showing up half or less.

File	Symbol	Code Snippet
omdurman-rules/src/board_data.rs:37 GH:omdurman-rules/src/board_data.rs:37	fall_of_khartoum_map	<pre>35 36 /// The Fall-of-Khartoum board (§9.3). 37 pub fn fall_of_khartoum_map_data() -> MapData { 38 FOK.get_or_init(parse(FOK_ROM)).clone() 39 }</pre>

Covered by tests: omdurman-rules::src::effects::fall_of_khartoum_board_excludes_no_hexes

§9.343 – Both players use the Dervish Range Effects Table (Bonus)

implemented

manual page unknown

Both players must use the Dervish Range Effects Table.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:3418 GH:omdurman-rules/src/effects.rs:3418	range_band_for	<pre>3416 /// in FALL OF KHARTOUM *both* players use the Dervish Range Effects Table 3417 /// (§9.343). 3418 pub fn range_band_for(3419 scenario: Scenario, 3420 player: Player,</pre>

Covered by tests: omdurman-rules::src::effects::fok_both_players_use_dervish_range_table

§9.344 – Dervish controls the North Fort (Bonus)

implemented

manual page unknown

The Dervish player controls the “North Fort” and may fire its guns.

File	Symbol	Code Snippet
omdurman-rules/src/scenario_setup.rs:42 GH:omdurman-rules/src/scenario_setup.rs:42	FALL_OF_KHARTOUM	<pre>40 /// North Fort uses a campaign HadendowaForts counter 41 (one of the spare fort 42 /// sprites). 43 pub const FALL_OF_KHARTOUM_SETUP: &[FixedPlacement] = 44 &[45 FixedPlacement { 46 section: SectionName::BritishBoats,</pre>
omdurman-rules/src/effects.rs:2205 GH:omdurman-rules/src/effects.rs:2205	hex_has_enemy_fort	<pre>2203 /// may neither occupy an enemy fort nor advance 2204 after combat into one 2205 /// (forts are never captured -- only destroyed, 2206 \$6.62/\$6.53/\$7.6). 2207 pub fn hex_has_enemy_fort(&self, hex: HexCoord, 2208 mover: Player) -> bool { 2209 self.units.iter().any(u { 2210 u.position == hex</pre>

Covered by tests: [omdurman-app::src::scenario_setup::fall_of_khartoum_places_gordon_in_the_palace](#)
[omdurman-app::src::scenario_setup::fall_of_khartoum_fort_landmarks_sit_at_the_correct_hexes](#)
[omdurman-app::src::scenario_setup::placement_done_gate_matches_by_identity_not_allocated_id](#)
[omdurman-rules::src::effects::fok_order_of_battle_dervish](#)
[omdurman-types::src::lib::fok_picker_allowlist_has_dervish_entry_force_blocks](#)

§9.345 – Gunboat White Nile <-> Blue Nile crossing (Bonus)

implemented

manual page unknown

The British gunboats may move from the White Nile to the Blue Nile and vice-versa at an off-board movement cost of six “upstream” movement points.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:2189 GH:omdurman-rules/src/effects.rs:2189	is_nile_mouth_crossing	<pre>2187 /// must be named on the board, else this is 2188 `false` and the move falls 2189 /// through to the ordinary contiguous-Nile rules. 2190 pub fn is_nile_mouth_crossing(&self, from: 2191 HexCoord, to: HexCoord) -> bool { 2192 let white = self 2193 .board</pre>
omdurman-types/src/lib.rs:518 GH:omdurman-types/src/lib.rs:518	Location::WhiteNileMouth	<pre>516 /// The off-board mouth of the White Nile branch 517 (FALL OF KHARTOUM §9.345) -- 518 /// a British gunboat may cross to the Blue Nile 519 mouth for 6 upstream MP. 520 WhiteNileMouth, 521 /// The off-board mouth of the Blue Nile branch 522 (FALL OF KHARTOUM §9.345). 523 BlueNileMouth,</pre>

Covered by tests: [omdurman-rules::src::effects::fok_gunboat_crosses_between_nile_mouths](#)

§9.346 – GORDON immobile, eliminated only at the Palace (Bonus)

implemented

manual page unknown

The GORDON leader unit starts in the palace and may not move during the scenario. He may only be eliminated by a Dervish unit passing through or occupying the palace hex (as normal movement or as advance after combat).

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:3114 GH:omdurman-rules/src/effects.rs:3114	check_gordon_palace	<pre>3112 /// after combat). Records the turn (which fixes the 3113 /// ends the game. A no-op outside FoK, or once GORDON 3114 pub fn check_gordon_palace(state: &mut GameState) { 3115 if state.scenario != Scenario::FallOfKhartoum 3116 state.gordon_eliminated_turn.is_some() { 3117 return;</pre>
omdurman-rules/src/lib.rs:591 GH:omdurman-rules/src/lib.rs:591	UnitIdentity::is_gordon	<pre>589 /// Whether this is the GORDON leader unit (§9.32, 590 §9.346) -- the immobile 591 /// palace defender whose elimination ends FALL OF 592 KHARTOUM (§9.35). 593 pub fn is_gordon(&self) -> bool { 594 matches!(595 self,</pre>

Covered by tests: [omdurman-rules::src::unit_profiles::gordon_is_an_immobile_british_leader](#)
[omdurman-rules::src::effects::gordon_survives_means_no_elimination](#)
[omdurman-app::src::scenario_setup::fall_of_khartoum_places_gordon_in_the_palace](#)

 7/10 implemented

§10 – Optional Rules

§10 – Optional Rules

descriptive

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Optional Rules (Campaign game only)

It is suggested that the most intriguing employment of the following two options is to permit the Dervish player to have either one or the other, but the Anglo-Egyptian player doesn't know which one until he stumbles onto it. Players are advised that the employment of both optionals in the same game is not recommended.

§10.1 – River Mines

descriptive

manual page unknown

River Mines

The Khalifa twice tried (unsuccessfully) to submerge a powerful mine in the Nile to sink or damage British gunboats. This option assumes that both attempts were successful.

§10.2 – River Chain

descriptive

manual page unknown

River Chain

The Khalifa also tried (also unsuccessfully) to string a heavy chain across the Nile to stop or slow down the British gunboats. This option assumes the chain was emplaced.

§10.11 – Secretly record mine hexes

implemented

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Prior to the commencement of play the Dervish player secretly records two Nile River hexes to be mined (the mines may not both be placed in the same hex). These hexes must be south of the E-W hexrow in which the Khor Shambat empties into the Nile.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:5217 GH:omdurman-rules/src/effects.rs:5217	apply_place_mine	<pre>5215 /// Lay a river mine during setup (§10.11). Validated by 5216 /// [<code>GameState::can_place_mine`</code>]. 5217 pub fn apply_place_mine(state: &mut GameState, hex: HexCoord) -> Result<(), RuleError> { 5218 state.can_place_mine(hex)?; 5219 state.mines.push(MinePlacement {</pre>
omdurman-rules/src/effects.rs:7768 GH:omdurman-rules/src/effects.rs:7768	GameState::mines	<pre>7766 #[rulebook("§10.11", "§10.21")] 7767 #[test] 7768 fn mines_and_chain_require_their_optional_rule() { 7769 // Without the optional rules selected, placement is rejected even in 7770 // Setup with room to spare.</pre>
omdurman-rules/src/	OptionalRule	<pre>300 /// two should be in play (rulebook §10). 301 #[derive(Serialize, Deserialize, Clone, Copy,</pre>

lib.rs:302 GH:omdurman-rules/src/lib.rs:302		<pre>PartialEq, Eq, Debug)] 302 pub enum OptionalRule { 303 RiverMines, 304 RiverChain,</pre>
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Covered by tests: [omdurman-rules::src::effects::mine_and_chain_limits_enforced_in_setup](#)
[omdurman-rules::src::effects::mines_and_chain_require_their_optional_rule](#)

§10.12 – Mine resolution

implemented

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When a British gunboat enters a mined hex, the Dervish player must order it to stop as it has struck a mine. The Dervish player then resolves the effect of the mine’s blast by rolling the ten-sided die:

- 1–4: No effect
- 5–7: Gunboat damaged, lost use of its engines and must drift two hexes per turn (with the current) for the rest of the game. No effect on guns or Maxims unless they drift out of range.
- 8–10: Gunboat sunk!

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:233 GH:omdurman-rules/src/effects.rs:233	RiverMine	<pre>231 // -- Optional rules ----- 232 /// River mine resolution (rulebook §10.12). 233 RiverMine { 234 gunboat_id: UnitId, 235 hex: HexCoord,</pre>
omdurman-rules/src/effects.rs:5132 GH:omdurman-rules/src/effects.rs:5132	apply_river_mine	<pre>5130 5131 /// Apply a river-mine resolution (rulebook §10.12). 5132 pub fn apply_river_mine(5133 state: &mut GameState, 5134 gunboat_id: UnitId,</pre>
omdurman-rules/src/lib.rs:1035 GH:omdurman-rules/src/lib.rs:1035	MineResult	<pre>1033 /// British gunboat enters a mined hex. 1034 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1035 pub enum MineResult { 1036 /// Roll 1-4: no effect. 1037 NoEffect,</pre>

Covered by tests: [omdurman-rules::src::effects::mine_fires_once_and_spare_s_dervish](#)

§10.13 – Mines consumed after both rolled for

implemented

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After both mines have been rolled for, no more are available.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:5132 GH:omdurman-rules/src/effects.rs:5132	apply_river_mine	<pre>5130 5131 /// Apply a river-mine resolution (rulebook §10.12). 5132 pub fn apply_river_mine(5133 state: &mut GameState, 5134 gunboat_id: UnitId,</pre>

Covered by tests: `omdurman-rules::src::effects::mine_fires_once_and_spares_dervish`

§10.14 – Dervish gunboats pass safely

implemented

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The Dervish player's gunboats may pass through the mined hexes with no ill effect (he knows where the mines are).

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:5132 GH:omdurman-rules/src/effects.rs:5132	apply_river_mine	<pre>5130 5131 /// Apply a river-mine resolution (rulebook §10.12). 5132 pub fn apply_river_mine(5133 state: &mut GameState, 5134 gunboat_id: UnitId,</pre>

Covered by tests: `omdurman-rules::src::effects::mine_fires_once_and_spares_dervish`

§10.21 – Secretly record chain hexes

implemented

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Prior to the commencement of play the Dervish player secretly records a line of river hexes (not exceeding four hexes long) across which the chain is strung. The hexes must be south of the E–W hexrow in which the Khor Shambat empties into the Nile.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1436 GH:omdurman-rules/src/effects.rs:1436	can_place_chain	<pre>1434 /// Read-only check of a river-chain placement in 1435 setup (§10.21): Setup phase 1436 /// and at most [<code>MAX_CHAIN_HEXES</code>] hexes. 1437 pub fn can_place_chain(&self, hexes: &[HexCoord]) 1438 -> Result<(), RuleError> { 1439 self.require_setup_phase()?; 1440 // Optional-rule gate: the chain exists only 1441 when the River Chain option</pre>
omdurman-rules/src/effects.rs:5228 GH:omdurman-rules/src/effects.rs:5228	apply_place_chain	<pre>5226 /// Lay (or replace) the river chain during setup 5227 (§10.21). Validated by 5228 /// [<code>GameState::can_place_chain</code>]. 5229 pub fn apply_place_chain(state: &mut GameState, hexes: 5230 &[HexCoord]) -> Result<(), RuleError> { 5231 state.can_place_chain(hexes)?; 5232 state.chain = Some(ChainPlacement {</pre>
omdurman-rules/src/effects.rs:4003 GH:omdurman-rules/src/effects.rs:4003	MAX_CHAIN_HEXES	<pre>4001 4002 /// Maximum contiguous Nile hexes the river chain may 4003 span (§10.21). 4004 pub const MAX_CHAIN_HEXES: usize = 4; 4005 -----</pre>

Covered by tests: `omdurman-rules::src::effects::mine_and_chain_limits_enforced_in_setup`
`omdurman-rules::src::effects::mines_and_chain_require_their_optional_rule`

§10.22 – Gunboat stops on chained hex

implemented

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When a British gunboat enters a “chained” river hex it must stop and may move no further that turn.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1705 GH:omdurman-rules/src/effects.rs:1705	can_move_gunboat	<pre>1703 /// upstream movement allowance is their maximum for that turn." Chained Nile 1704 /// hexes stop the gunboat (§10.22). 1705 pub fn can_move_gunboat(&self, 1706 unit_id: UnitId,</pre>

Covered by tests: [omdurman-rules::src::effects::chain_stops_gunboat_until_sunk](#)

§10.23 – Sinking the chain

implemented

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No gunboats (British or Dervish) may cross the chain until it has been sunk by the British player. He may sink the chain by a) having an infantry or cavalry unit spend one complete turn on either riverbank adjacent to a “chained” river hex, or b) firing at the chain with artillery and achieving a 3 or more on the Combat Results Table.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:5176 GH:omdurman-rules/src/effects.rs:5176	apply_sink_chain	<pre>5174 /// Sink the river chain (rulebook §10.23). Marks the placed chain cleared so it 5175 /// no longer stops gunboats (§10.22). 5176 pub fn apply_sink_chain(state: &mut GameState) -> Result<(), RuleError> { 5177 match state.chain.as_mut() { 5178 Some(chain) if !chain.sunk => {</pre>

Covered by tests: [omdurman-rules::src::effects::chain_stops_gunboat_until_sunk](#)

0/1 implemented

§11 – Historical Notes

§11 – Historical Notes

descriptive

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Historical Notes

In 1881 Mohammed Ahmed Ibn Al-Sayid Abdullah, the son of an obscure carpenter in the hinterlands of the Sudan, proclaimed himself the “Mahdi” — the Messiah of the Islamic faith. His timing was propitious indeed. Since the early 1820’s a corrupt Egypt, with the Sultan of Turkey’s blessing, had incessantly raped the Sudan, taking ivory and flooding the slave markets with some half million captured Sudanese blacks. By 1880, nearly 40,000 Egyptian troops occupied outposts scattered throughout the Sudan, enforcing Egypt’s hold on this lucrative ivory and slave trade and squeezing the native population dry through vicious and corrupt tax officials. All was controlled from Khartoum via the office of Governor General of the Sudan. The title had been held by a succession of individuals, including General Charles Gordon, whose appointment was an attempt to reinstate some rudimentary justice in the Sudan after France and Britain assumed joint political control of a bankrupt Egypt.

By 1881, however, Gordon’s term had expired and a new Governor General, again corrupt and incompetent,
(see manual for full text)

See also: §9.1, §9.2, §9.3

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Credits

§Credits – Credits

descriptive

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Combat Results Table (shared reference)

\$CRT – Combat Results Table (shared by \$6.22 fire and \$7.7 melee)

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File	Symbol	Code Snippet
omdurman-rules/src/combat_results_table.rs:8 GH:omdurman-rules/src/combat_results_table.rs:8	FireFactorRow	<pre>6 /// to index into the result matrix. 7 #[derive(serde::Serialize, serde::Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 8 pub enum FireFactorRow { 9 /// 1-5 factors 10 Row01to05,</pre>
omdurman-rules/src/combat_results_table.rs:44 GH:omdurman-rules/src/combat_results_table.rs:44	from_total	<pre>42 43 /// Determine which row a given total fire factor falls into (rulebook \$6.22). 44 pub fn from_total(total: u16) -> Self { 45 match total { 46 0..=5 => FireFactorRow::Row01to05,</pre>
omdurman-rules/src/combat_results_table.rs:85 GH:omdurman-rules/src/combat_results_table.rs:85	combat_results_table	<pre>83 /// D = `Disrupt` (1/2 of target units, round up) 84 /// 1..5 = `Eliminate(n)` (that many units removed) 85 pub fn combat_results_table(row: FireFactorRow, roll: DieRoll) -> CombatResult { 86 let cells = crate::tables_data::crt_table() 87 .get(&row)</pre>
omdurman-rules/src/lib.rs:923 GH:omdurman-rules/src/lib.rs:923	CombatResult	<pre>921 /// * `` -- no effect 922 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 923 pub enum CombatResult { 924 NoEffect, 925 Disrupt,</pre>

Covered by tests: [omdurman-rules::src::combat_results_table::ae_combat_results_table_lowest_is_no_effect](#)

[omdurman-rules::src::combat_results_table::ae_combat_results_table_highest_is_eliminate_5](#)

[omdurman-rules::src::combat_results_table::ae_combat_results_table_progresses_with_roll](#)

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[omdurman-rules::src::combat_results_table::crt_all_rows_monotone_non_decreasing](#)

[omdurman-rules::src::combat_results_table::crt_cross_row_monotone_for_each_roll](#)

[omdurman-rules::src::combat_results_table::crt_lowest_row_is_worst_highest_row_is_best](#)

[omdurman-rules::src::combat_results_table::crt_eliminate_never_exceeds_5](#)

[omdurman-rules::src::combat_results_table::crt_every_cell_matches_the_table](#)

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| Reference – Charts and Tables

§Reference – Charts and Tables

out-of-scope

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